

Narrow Inference and Incentive Design^{*}

Alexander Clyde[†]

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Abstract

There is evidence that people struggle to infer how their choices map into outcomes when faced with complex multidimensional choice problems. This paper explores the consequences of this in a principal-agent model. A principal chooses a mechanism to sort different types of agents into choosing different action combinations. The agents make choices on multiple dimensions, and infer the effect of each action separately without properly controlling for the other actions. I characterize the principal’s optimal mechanism when facing agents who do such ‘narrow’ inference, and contrast it with their optimal mechanism when the agents are fully rational.

Keywords: Behavioural Mechanism Design, Screening, Narrow Bracketing, Misspecified Models.

JEL Classification: D90, D02, D82

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[†]Department of Economics, Aalto University School of Business; Address: Ekonominaukio 1, FI-02150 Espoo, Finland; Email: alexander.clyde.econ@gmail.com or alexander.clyde@aalto.fi

1 Introduction

The literature on incentive design in economics commonly assumes that people face a well-understood schedule mapping their choices into outcomes. However, in reality economic decision makers do not always face an explicit posted schedule. Large organizations do not post a wage function detailing how every combination of skill, task and occupation maps into career earnings. Nevertheless, they do shape features of the institutional environment, such as promotion policy, compensation structure, internal labour markets and task organization. Such institutional environments induce mappings from multidimensional actions into outcomes that are complex and opaque. A body of economic literature suggests that people often struggle to correctly understand such mappings.

In this paper, I study incentive design under the assumption that people form beliefs about mappings from multidimensional action choices to outcomes by observing data generated by others facing the same choices. I assume they do so in a way that treats each dimension of decision narrowly and separately, according to a procedure I call *narrow inference*. For each decision, they compare the average outcome of those who take an action to the average outcome of those who do not. In forming beliefs about the effect of any individual action in this way, they fail to control for other dimensions of action. This results in a distortion of their beliefs that is similar to the omitted variable bias that would arise if a researcher ran a separate univariate regression to estimate the effect of each of a series of correlated treatments.

I explore how a principal would design an incentive mechanism if they knew the agents they interact with had this form of bounded rationality. I obtain results characterizing the principal's optimal mechanism with agents who make narrow inference under two different participation constraint regimes. The results are useful both because they provide a recipe for how to solve particular parameterized applications of the incentive design problem under narrow inference, and because they allow us to make clear comparisons between the principal's optimal mechanism under narrow inference and the principal's optimal mechanism when agents fully understand an outcome mapping.

I build on the characterization results to show how narrow inference changes the shape of the optimal mechanism. I show that this fundamentally depends on the nature of the contracting relation between the principal and the agents. To see this, consider the following two applications. In the first, the principal is a large organization employing the agents as workers. The agents have to infer how

different dimensions of skill they can acquire will affect their career earnings within the organization. The principal shapes the internal labour markets, promotion and compensation structure which determine how these skills will map into ultimate earnings. In the absence of a posted wage schedule, the workers have to infer the relationship between skills and earnings from any available data. This is in line with empirical evidence from labour economics. [Backhaus et al. \(2023\)](#) find evidence that workers read raw pay gaps between part-time and full-time work as the causal impact of switching to full-time work. [Jensen \(2010\)](#) and [Wiswall and Zafar \(2015\)](#) both present evidence that students form beliefs about the earnings returns to schooling choices using observed population outcomes.

In the second application, the principal is a professional services firm shaping a billing architecture for the agents, their clients. This is a complex price structure for the multiple service lines the client can choose from. The client cannot obtain a clear binding price quote before the firm has already been contracted and determined the size and boundaries of the service being delivered. In order to estimate the cost of each service, the client has to obtain data on what comparable clients actually ended up paying. They can obtain this data from a variety of sources, such as benchmarking surveys, procurement consultants or peer networks. There is a substantial literature suggesting that, faced with such complex price schemes, people track coarse aggregate statistics of realized payments rather than the true schedule ([Ito, 2014](#); [De Bartolome, 1995](#); [Liebman and Zeckhauser, 2004](#); [Rees-Jones and Taubinsky, 2020](#); [Feldman et al., 2016](#))— even when schedules are fully posted. Similarly, evidence from insurance plan choices ([Abaluck and Gruber, 2011](#); [Bhargava et al., 2017](#)) suggests that faced with multidimensional choices people can struggle to understand price schedules even when they are fully disclosed.

In the first application the principal is an employer who benefits from the agents taking the costly actions to acquire these skills. I show that in settings like these the principal unambiguously gains, relative to the rational benchmark and responds by inducing a greater proportion of workers to acquire each skill. How far this extends across skills depends on the participation constraint regime, which I make precise in Section 4. In contrast, in settings like the second application where the principal is providing costly services benefiting the agent, narrow inference hurts the principal and leads them to induce fewer types of agent to take the actions. Finally, I show that under narrow inference the principal can benefit from bundling actions in the second application but not the first. In Section 5.1, I present evidence that fits this pattern.

In the formal model, the agents face a binary decision problem on whether to take an action or not on multiple dimensions. The principal designs an institutional environment inducing a mapping from actions to a transfer. This is formally modelled as the principal committing to a transfer mapping t . There is a large population of agents who differ according to a single-dimensional type variable that affects the predictable costs and benefits of the actions. The utility of the agents can be decomposed into a type-dependent predictable part that they fully understand, and a transfer part that they need to infer. Each agent’s type is private information, and the principal’s choice of transfer mapping screens different types of agents into choosing different actions. With fully rational agents who know and understand the true transfer mapping this is a standard monopolistic screening problem. Throughout, the transfer mapping should be read as the reduced form of an institutional environment — a compensation policy, billing architecture, or platform design — mapping multidimensional actions into realized outcomes. Agents are uncertain not about a rule that could be disclosed, but about the outcomes the institution generates, and they estimate these from population data.

My analysis of the design problem proceeds as follows. First, in order to contrast the principal’s optimal mechanism when agents make narrow inference to that with fully rational agents, I state results adapted from [Carroll \(2017\)](#) which describe the principal’s optimal mechanism for the rational benchmark. Under a standard regularity assumption, the principal’s optimal mechanism is fully separable across dimensions. Facing such a mechanism, the agents choose a strategy that on each dimension selects the action if and only if their type is above a dimension-specific threshold.

I then describe the two results characterizing the principal’s optimal mechanism under narrow inference under two different participation constraint regimes. In both results, the principal can solve for their optimal mechanism under narrow inference by maximizing an expression analogous to the one that yields their optimal mechanism in the rational benchmark. The difference is that the part of the objective corresponding to agents’ predictable utilities is shrunk towards zero by some constant factors that lie between one and zero, with the shrinkage factors summing to one across dimensions. Under belief-based ‘narrow participation constraints’, the characterization result is as if there is an adversarial player choosing these shrinkage factors to minimize the objective of the principal in a zero-sum game. The results demonstrate how narrow inference amplifies the perceived size of transfers relative to their actual size. The difference between agents’ beliefs under narrow and rational inference is driven by

the fact that the equilibrium distribution over actions is correlated across dimensions, something that is possible due to joint dependence on the type variable.

Thus, the size of the transfers required to implement any perceived size of incentives is reduced under narrow inference. This means that how the welfare of the principal and the equilibrium strategy of the agents change under narrow inference relative to the rational benchmark depends on whether the agents are a recipient or a contributor of transfers in equilibrium. This is what drives the results in settings like those described in the two applications.

The paper takes a step towards understanding how errors in causal inference might affect how we should design economic incentives by studying a novel model of belief formation. In Section 5.2, I discuss the key assumptions of the model and how they could be relaxed in future work building on this paper.

In what follows, I develop the employer-worker application to illustrate narrow inference and to demonstrate how the principal might benefit from agents using narrow inference.

Example 1. The agents have two dimensions of action, whether to obtain technical skills $a_1 = 1$ or not $a_1 = 0$ and whether to obtain managerial skills $a_2 = 1$ or not $a_2 = 0$. The principal, an employer, chooses an institutional environment that induces an earnings mapping within the organization that depends jointly on these two actions $t : A_1 \times A_2 \rightarrow \mathbb{R}$. There are three types of agent $s \in \{0, 1, 2\}$. The probabilities of the types are denoted $p_0, p_1, p_2 \in (0, 1)$ respectively. The agents' utility depends on their type s , their actions a_1, a_2 and the transfer t .

$$t(a_1, a_2) - (3 - s)(a_1 + a_2)$$

Where $-(3 - s)(a_1 + a_2)$ is the predictable utility cost of making human capital investments for the worker. Suppose that the principal wants to implement that the type $s = 0$ chooses neither action, the type $s = 1$ obtains technical skills $a_1 = 1$ but not managerial skills $a_2 = 0$, while the highest type $s = 2$ obtains both $a_1 = a_2 = 1$. An earnings function that ensures that rational agents will act this way must satisfy the following incentive constraints. The first ensures that type $s = 1$ chooses $(1, 0)$ over $(0, 0)$ and the second ensures type $s = 2$ chooses action $(1, 1)$ over $(1, 0)$.

$$\begin{aligned} t(1, 0) - 2 &\geq t(0, 0) \\ t(1, 1) - 2 &\geq t(1, 0) - 1 \end{aligned}$$

Choosing t such that these incentive constraints bind allows the principal to minimize the earnings paid to types $s = 1$ and $s = 2$. This means $t(1, 1) > t(1, 0) > t(0, 0)$. These local incentive constraints binding suffice for all incentive constraints to hold.

Now consider what happens if the agents use narrow inference, but the transfer mapping is fixed at values that make the rational incentive constraints bind. They expect the earnings from any action to be the average-population level earnings of those who have taken that action. For an agent of type $s = 1$ making narrow inference to want to invest in technical skills requires that

$$\frac{p_1}{p_1 + p_2}t(1, 0) + \frac{p_2}{p_1 + p_2}t(1, 1) - 2 \geq t(0, 0)$$

Their narrow beliefs about the earnings from acquiring technical skills pool the earnings of both the types who obtain these skills, who have proportion $p_1 + p_2$ in the population of types. The narrow beliefs about the earnings from not acquiring technical skills are in line with rational beliefs, since only the lowest type $s = 0$ doesn't acquire these skills. Since $t(1, 1) > t(1, 0)$, the narrow perception of the expected earnings benefit of technical skills is biased upward from the true effect. It fails to adjust for the fact that a type $s = 1$ agent who is on the margin between obtaining technical skills does not obtain managerial skills and thus has lower future earnings than the average worker who obtains technical skills.

Similarly, the narrow perception of the earnings benefit of obtaining managerial skills is biased upward. This is because the expected earnings of those without managerial skills pools types who did and did not obtain technical skills, while the expected wage of those with managerial skills coincides with the true earnings. This upward bias in the incentives the agents perceive on both dimensions allows the principal to reducing earnings across the type distribution while inducing that on each dimension at least as many types take the actions 1 as before. $\triangle$

Literature Review

Empirical work in economics and psychology documents people making inferential errors similar to narrow inference. Several papers find evidence of people estimating causal effects using one-dimensional conditional averages. In [Backhaus et al. \(2023\)](#), workers are found to form expectations about the causal effect of full-time vs part-time work using data on the pay gap between the two that has not been adjusted for individual worker characteristics and other choices. In experimental work by [Barron et al. \(2024\)](#) subjects infer expected investment returns from historical data on oth-

ers' decisions, without accounting for the selection process driving those decisions. Graeber (2023), Fernbach et al. (2010) and He and Kučinskas (2024) all find a tendency for experimental subjects to narrowly focus on a few variables when making causal predictions, neglecting other variables that may contain valuable information.

Narrow inference involves causal misperceptions, but also thinking about decision problems narrowly. The literature on narrow bracketing considers decision makers who break decision problems into smaller sub-problems without accounting for how these decisions interact in the larger joint problem. Work in this area: Tversky and Kahneman (1981), Thaler (1985), Thaler (1999), Read et al. (1999), Rabin and Weizsäcker (2009), has both documented evidence for and explored the theoretical implications of narrow decision making. Recent work has explored theoretical foundations for narrow bracketing based on costly information acquisition (Kőszegi and Matějka, 2020), information frictions (Lian, 2021) and computability constraints (Camara, 2022). Vorjohann (2026) presents an axiomatization reconciling tensions between narrow decision making and global budget balance.

In modelling agents with narrow causal perceptions, I build on work studying decision making by agents who have misspecified models of how action choices map into consequences. There is a growing literature on the Berk-Nash Equilibrium of Esponda and Pouzo (2016), a solution concept founded as the limit of a process of misspecified learning; (Heidhues et al., 2018; Frick et al., 2020; Bohren and Hauser, 2021; Fudenberg et al., 2021). Frick et al. (2022) propose a related concept, Assortativity Neglect Equilibrium, that models incorrect inference applied to data observed from the rest of the population. This is similar to narrow inference, which is an inferential error by agents using data on the population of agents interacting with the mechanism. Another connected literature is that on modelling causal misperceptions using Bayesian Networks; (Spiegler, 2016; Eliaz and Spiegler, 2020). Schumacher and Thyssen (2022) use this Bayesian Network approach in a principal-agent moral hazard problem where the agent has causal misperceptions of how their actions map into output. In Eliaz and Spiegler (2026) a Bayesian Network formalism is used to model the design of narratives for misspecified news consumers by media organizations.

Earlier work on design when agents misperceive incentives by Rubinstein (1993) and Piccione and Rubinstein (2003) explores monopolistic pricing when customers have a coarse misperception of any pricing strategy. Eyster and Rabin (2005) consider a solution concept for games where players neglect correlation between information and opposing players' actions, and apply their concept to bilateral trade and auction settings. Jehiel (2005) develops an equilibrium concept for extensive-form games

—Analogy Based Expectation Equilibrium (ABEE)— in which players have coarse misperceptions of other players’ strategies.¹ The first papers to explicitly apply the ABEE concept to design problems are [Jehiel \(2011\)](#) and [Jehiel and Mierendorff \(2024\)](#) who study auction design where bidders receive limited feedback. A paper explicitly studying narrow bracketing and auction design is [Eisenhuth \(2019\)](#), who uses a prospect theoretic formulation of narrow bracketing.

In contributing to the small literature on mechanism design where agents use misspecified models, this paper also contributes to a larger literature on mechanism design that takes into account agents’ limited rationality in a variety of other dimensions. A detailed review can be found in [Kőszegi \(2014\)](#). This includes work in contract theory ([Eliaz and Spiegel, 2006](#)), ([Heidhues and Kőszegi, 2010](#)), ([Herweg et al., 2010](#)) and optimal taxation ([O’Donoghue and Rabin, 2006](#)), ([Spinnewijn, 2015](#)), ([Farhi and Gabaix, 2020](#)), ([Lockwood, 2020](#)). Some models of optimal income taxation under behavioural biases are built on empirical findings that people have coarse misperceptions of income tax schedules that are similar to narrow inference; ([Liebman and Zeckhauser, 2004](#)), ([Feldman et al., 2016](#)), ([Rees-Jones and Taubinsky, 2020](#)).

2 Model

Each agent faces a multidimensional decision. Let $A = \{0, 1\}^n$ be the agents’ set of feasible action profiles. I refer to $i \in \{1, \dots, n\} \equiv N$ as a dimension, such that a_i is the agents’ action in dimension i . An agent has a type that lies in a bounded interval $s \in S \equiv [0, 1]$. This type is drawn from a distribution P that admits a density $p(s)$ such that $p(s) > 0$ for all $s \in (0, 1]$, and has an atom of probability $P(0) > 0$ at $s = 0$. Denote the cdf of the distribution by $P(s) = P(0) + \int_0^s p(\tilde{s})d\tilde{s}$. This type is the agents’ private information and is unknown to the principal.

If the dimension i action a_i is taken by a type s agent it generates predictable utilities $v_i(s)a_i$ and $w_i(s)a_i$ for the agent and the principal respectively. The functions $v_i(s)$ and $w_i(s)$ are continuously differentiable on $s \in (0, 1]$, with well-defined limits $v_i(0) \equiv \lim_{s \downarrow 0} v_i(s)$, $w_i(0) \equiv \lim_{s \downarrow 0} w_i(s)$. We assume $v_i(s)$ is strictly increasing on all dimensions while $w_i(s)$ is weakly increasing on all dimensions. Type $s = 0$ is an *inactive type* who always chooses the action 0 on all dimensions.² This is a mass of

¹The behaviour of the principal and the agents under narrow inference can be formulated as an ABEE of an extensive-form game, and I discuss this in more detail in Online Appendix Section [A.1](#).

²The assumption there is an atom of probability at $s = 0$ ensures that the action combination $a = 0$ has a positive mass of types taking it.

agents who never engage with the mechanism: workers who acquire no skills, clients who commission no service lines. Since narrow beliefs are estimated from population data, these agents matter under narrow inference even though they can be ignored in the rational benchmark.

In addition to the predictable utilities, the agent receives utility from a transfer t that needs to be inferred. The utility of an agent of type s , choosing action combination $a = (a_i)_{i \in N}$ with transfer $t \in \mathbb{R}$ is

$$u(s, a, t) = \sum_{i \in N} v_i(s) a_i + t \quad (1)$$

The transfer $t \in \mathbb{R}$ represents a zero-sum transfer of surplus between the agent and the principal. The principal's payoff given that actions a are taken by a type s agent, resulting in transfer $t \in \mathbb{R}$ is

$$W(s, a, t) = -t + \sum_{i \in N} w_i(s) a_i \quad (2)$$

Throughout the paper, I make the following standard regularity assumption on the type distribution.³

Assumption 1. For every dimension $i \in N$

$$\phi_i(s) = v_i(s) - \frac{1 - P(s)}{p(s)} v'_i(s)$$

is strictly increasing in $s \in (0, 1]$. We refer to this property as *increasing virtual values* (IVV).

Integrating by parts, for every dimension $i \in N$ and every threshold $\hat{s}_i \in [0, 1]$, $V_i(\hat{s}_i) \equiv \int_{\hat{s}_i}^1 \phi_i(s) p(s) ds = v_i(\hat{s}_i)(1 - P(\hat{s}_i))$, where at $\hat{s}_i = 0$ the right-hand side is evaluated using $v_i(0) \equiv \lim_{s \downarrow 0} v_i(s)$, so that $V_i(0) = v_i(0)(1 - P(0))$ is finite. Thus V_i is unaffected by the atom at $s = 0$ except through the factor $1 - P(\hat{s}_i)$. I write $V_i(\hat{s}_i)$ for this quantity throughout.

2.1 Mechanisms

I focus on a natural class of indirect mechanisms. Before the agents take any actions, the principal commits to a mechanism, which consists of a function mapping actions

³In Section A.6 of the Online Appendix, I explore the implications of relaxing this assumption.

to transfers $t : A \rightarrow \mathbb{R}$.⁴ After learning their type, each agent chooses a distribution over actions according to a strategy $g : S \rightarrow \Delta(A)$. The marginal distribution over actions in dimension i is denoted by $g_i(a_i|s) = \sum_{a_{-i} \in A_{-i}} g(a_i, a_{-i}|s)$.

For a transfer mapping $t \in \mathbb{R}^A$, given a strategy g the expected payoff for the principal is

$$W(t, g) = \int_0^1 \sum_{a \in A} [-t(a) + \sum_{i \in N} w_i(s) a_i] g(a|s) dP(s) \quad (3)$$

2.2 Model Interpretations

The model allows actions to have both a positive and negative effect on payoffs. The sign of $v_i(s)$ determines whether a type s agent has predictable positive utility from action $a_i = 1$ or predictable disutility. Likewise, the direct effect of actions on the principal's payoff can be positive ($w_i(s) \geq 0$) or negative ($w_i(s) \leq 0$).

Take the applications suggested in the introduction. In the first, the agents are workers and the principal a large employing organization. Here $v_i(s) < 0$ is the cost of obtaining human capital and $w_i(s) > 0$ is the benefit to the organization of the human capital the worker acquires. The organization determines the compensation structure, promotion policy, and internal labour market which induce a mapping between skills and the transfer $t : A \rightarrow \mathbb{R}$.

In the second application the agent is a client purchasing services from the principal, who is a professional services firm. The agents derive positive utility $v_i(s) > 0$ from these services, but they are costly for the principal to provide $w_i(s) < 0$. The principal's choice of their billing architecture induces a mapping from the clients' choice of service lines into the transfer $t : A \rightarrow \mathbb{R}$. The agents' type affects both their own utility from the services and the principal's costs of providing them.

The model best fits a population interpretation, where the principal sets a mechanism and then repeatedly interacts with a population of short-run agents with types distributed according to $P(s)$. The equilibrium under both narrow inference and the rational benchmark is then a steady state of this process.

⁴We can incorporate uncertainty into the transfer mapping through a random additional variable $\omega \in \Omega \subseteq \mathbb{R}$, which is finite with distribution μ . The transfer mapping is then $t : A \times \Omega \rightarrow \mathbb{R}$. We can perform all the analysis of this paper with the expected transfer mapping $t(a) = \sum_{\omega \in \Omega} t(a, \omega) \mu(\omega)$, since the transfer enters linearly in the utilities of the principal and the agents.

2.3 Rational Inference

Given a strategy g , write the expected utility of an agent of type s as

$$U(s) = \sum_{a \in A} g(a|s) u(s, a, t(a)) = \sum_{a \in A} g(a|s) \left[\sum_{i \in N} v_i(s) a_i + t(a) \right] \quad (4)$$

Incentive Compatibility (IC) of strategy g under transfer mapping $t \in \mathbb{R}^A$ requires that g is a best response to t . This means for any $s \in (0, 1]$, $a \in \text{supp}(g(\cdot|s))$ and any $a' \in A$

$$\sum_{i \in N} v_i(s) a_i + t(a) \geq \sum_{i \in N} v_i(s) a'_i + t(a') \quad (5)$$

An agent always has the option of not participating in the mechanism, taking the actions $a = 0$ and obtaining zero transfer. Thus the transfer conditional on action combination $a = 0$ must be at least zero. We call this the *limited liability at zero* constraint.⁵

$$t(0, \dots, 0) \geq 0 \quad (6)$$

Limited liability at zero ensures that an agent who takes no action cannot be charged.

2.4 Narrow Inference

Given a strategy g , an unconditional distribution over actions in A is induced as follows.

$$g(a) = \int_0^1 g(a|s) dP(s) \quad (7)$$

Let the marginal over an action in dimension i be denoted $g_i(a_i) = \sum_{a_{-i} \in A_{-i}} g(a_i, a_{-i})$.

An agent evaluates each dimension of action by the average realized transfer associated with that action in the population. Concretely, an agent believes when taking a decision in dimension i that in expectation they will receive $\bar{t}_i(a_i)$ if they take action a_i . When $g_i(a_i) > 0$ we require that this expectation is consistent with the actual conditional expectation of transfers given a_i .

⁵Together with the rational incentive constraints this is equivalent to a participation constraint requiring that all types obtain weakly positive expected utility: for all $s \in S$, $U(s) \geq 0$.

$$\bar{t}_i(a_i) = \sum_{a_{-i} \in A_{-i}} \frac{g(a_i, a_{-i})}{g_i(a_i)} t(a_i, a_{-i}) \quad (8)$$

Denote the vectors of beliefs $\bar{t}(a) = (\bar{t}_i(a_i))_{i=1}^n \in \mathbb{R}^n$ and $\bar{t} = (\bar{t}_i(0), \bar{t}_i(1))_{i \in N} \in \mathbb{R}^{2n}$. When $g_i(a_i) = 0$, we allow $\bar{t}_i(a_i)$ to take on arbitrary values. Analogously to the rational benchmark, these ‘off-path’ actions do not affect the principal’s objective and $\bar{t}$ can be set to ensure incentive compatibility. Henceforth, I refer to agents performing narrow inference as ‘narrow agents’.

A narrow agent imposes an additively separable form on their estimate of the transfer mapping using $\bar{t}$. A strategy g is *narrow incentive compatible* (NIC) if for any dimension $i \in N$, type $s \in (0, 1]$ and actions $a_i \in \text{supp}(g_i(\cdot|s))$, $a'_i \in A_i$

$$v_i(s)a_i + \bar{t}_i(a_i) \geq v_i(s)a'_i + \bar{t}_i(a'_i) \quad (9)$$

If a consistent transfer mapping and beliefs exist such that a strategy is NIC, we say that strategy is *NIC-implementable*. A mechanism that implements a NIC strategy is called an *NIC mechanism*.

I analyse the model under two forms of participation constraint, one weaker and one stronger. The first is the *limited liability at zero* constraint that was also considered in the rational benchmark: $t(0, \dots, 0) \geq 0$.⁶

The second form requires that, in addition to limited liability at zero, the mechanism must satisfy dimension-by-dimension *narrow participation constraints*. Under these constraints, for each dimension the expected transfer conditional on the action zero being taken on that dimension must be nonnegative. Formally, for all $i \in N$

$$\bar{t}_i(0) \geq 0 \quad (10)$$

The narrow participation constraints reflect the idea that the agent perceives each dimension as a separate contractual arrangement they can walk away from, for example that they have separate contracts for each task, and/or they are working according to gig-economy style arrangements. In contrast, the limited liability at zero constraint reflects legal or other restrictions on receiving transfer from an agent who has not taken any actions within the mechanism, such as an employee who has not undertaken any tasks or acquired any skills or a potential client who has not requested any service lines.

⁶The atom of types $s = 0$ who always choose the zero action on every dimension ensures that this constraint always matters under narrow inference.

Note that in the rational benchmark, the limited liability at zero constraint and participation constraints based on the agents' beliefs about the mechanism coincide. Under narrow inference, these two notions are distinct, and it is appropriate to analyse what additional restrictions belief-based participation constraints may put on the set of NIC mechanisms. We will see that adding the narrow participation constraints makes it easier to state conditions under which the principal attains an optimal mechanism (Corollary 1) and gives clearer comparative statics on how narrow inference changes the principal's optimal mechanism relative to the rational case (Proposition 4).

2.5 When do narrow incentive compatibility and rational incentive compatibility coincide?

If the true transfer mapping $t : A \rightarrow \mathbb{R}$ is additively separable across dimensions, and the action distribution is independent across dimensions, then narrow incentive compatibility coincides with rational incentive compatibility. To see this, rewrite the narrow belief consistency condition (8) under additive transfer mapping $t(a) = \sum_{i \in N} t^i(a_i)$ and an action distribution such that $g(a) = \prod_{i \in N} g_i(a_i)$ and $g_i(1) = q_i \in (0, 1)$ for all $i \in N$.

$$\begin{aligned} \bar{t}_i(a_i) &= \sum_{a_{-i} \in A_{-i}} \frac{g(a_i, a_{-i})}{g_i(a_i)} [t^i(a_i) + \sum_{j \in N \setminus \{i\}} t^j(a_j)] \\ &= t^i(a_i) + \sum_{a_{-i} \in A_{-i}} \frac{g(a_i, a_{-i})}{g_i(a_i)} \sum_{j \in N \setminus \{i\}} t^j(a_j) \\ &= t^i(a_i) + \sum_{a_{-i} \in A_{-i}} \prod_{j \in N \setminus \{i\}} q_j^{a_j} (1 - q_j)^{1-a_j} \sum_{j \in N \setminus \{i\}} t^j(a_j) \end{aligned} \quad (11)$$

The second term does not depend on $a_i \in A_i$, and it cancels through in the NIC condition, which for any $i \in N$, $s \in S$ and actions $a_i, a'_i \in A_i$ becomes

$$\bar{t}_i(a_i) - \bar{t}_i(a'_i) \geq v_i(s)a'_i - v_i(s)a_i \Leftrightarrow t^i(a_i) - t^i(a'_i) \geq v_i(s)a'_i - v_i(s)a_i$$

This is equivalent to rational IC under this additive transfer mapping.

3 Rational Benchmark

With rational agents, we have a screening problem with a single dimension of type but multiple dimensions of action. I restate existing results from [Carroll \(2017\)](#) adapted

to our setting.⁷

It will be shown that the agents' strategy under the principal's optimal mechanism with rational agents takes a *threshold form* where $g(a|s) = \prod_{i \in N} g_i(a_i|s)$ and there is a threshold $\hat{s}_i \in S$ on each dimension such that $g_i(1|s) = \mathbb{1}\{s \geq \hat{s}_i\}$ for all $s \in (0, 1]$ and $g_i(1|0) = 0$.⁸ We adopt the convention that $g_i(1|s) = 0$ for all $s \in S$ if $\hat{s}_i = 1$. Let the vector of thresholds across dimensions be denoted $\hat{s} = (\hat{s}_i)_{i \in N} \in [0, 1]^n$. We can characterize the principal's problem in terms of choosing these thresholds. Denote the value of the principal's objective under threshold strategy $\hat{s}$ by $W(\hat{s})$.

Proposition 1. *Assume the IVV assumption holds. The principal maximizes their objective over all IC mechanisms that satisfy the limited liability at zero constraint if and only if they choose a transfer mapping implementing a threshold strategy that solves the following problem.*

$$\max_{\hat{s} \in [0, 1]^n} W(\hat{s}) = \sum_{i \in N} \int_{\hat{s}_i}^1 (\Phi_i(s) + w_i(s)) p(s) ds \quad (12)$$

The principal's value under an objective-maximizing mechanism can be achieved by an additively separable transfer mapping

$$t(a_1, \dots, a_n) = \sum_{i \in N} t^i(a_i) \quad (13)$$

$$t^i(0) = 0, t^i(1) = -v_i(\hat{s}_i) \text{ for all } i \in N \quad (14)$$

Proof. [In Appendix](#) □

Thus, the principal's optimal transfer mapping can be treated as the sum of separate transfer mappings, one for each dimension. This does not result directly from IC, but rather from the optimality for the principal of implementing a threshold strategy when IVV holds.⁹

⁷In particular Proposition 3.1 of [Carroll \(2017\)](#). Earlier work by [Mirman and Sibley \(1980\)](#) analyses the monopoly screening model with many dimensions of action but a single dimension of type, but assumes directly that the IC action on any dimension is increasing in the type. In [Carroll \(2017\)](#) this is explicitly shown to hold under the regularity (IVV) assumption.

⁸Due to the atom of types $s = 0$ who choose $a = 0$ on every dimension.

⁹We see in Section [A.6](#) of the Online Appendix that without IVV it can be optimal for the principal to choose a non-separable transfer mapping, and that under such a transfer mapping a non-threshold strategy is implemented and thus IC.

4 Narrow Agents

The following definitions are used in the statement of the result. We say a threshold strategy $\hat{s} = (\hat{s}_i)_{i \in N}$ has no *coincident interior thresholds at distinct utilities* if there is no pair of distinct dimensions $i, j \in N$ with $\hat{s}_i = \hat{s}_j \in (0, 1)$ such that $v_i(\hat{s}_i) \neq v_j(\hat{s}_j)$. We say a threshold strategy $\hat{s} = (\hat{s}_i)_{i \in N}$ has no *coincident thresholds below one at distinct utilities* if there is no pair of distinct dimensions $i, j \in N$ with $\hat{s}_i = \hat{s}_j \in [0, 1)$ such that $v_i(\hat{s}_i) \neq v_j(\hat{s}_j)$.

We will see that any threshold strategy with coincident interior thresholds at distinct utilities cannot be made NIC by any transfer mapping. However, I show that any value that could only be attained by such a strategy can be approximated arbitrarily closely by a strategy with distinct thresholds across all dimensions.

The main results show that under narrow inference the principal faces the same design problem as in the rational benchmark except the agents' predictable utilities are scaled by some factor $\beta_i \in [0, 1]$ in each dimension. An interpretation of these scaling factors is as follows. For a given perceived size of incentives, you have a transfer mapping that would achieve these perceived incentives under narrow inference, and a transfer mapping that would achieve these perceived incentives under the rational benchmark. The scaling factors give by how much these transfers for the rational benchmark would have to be scaled to achieve the same expected transfer as under narrow inference. The fact the scaling factors must be between zero and one and sum to one demonstrates that narrow inference amplifies the size of incentives relative to the rational benchmark.

4.1 Main Characterization Results

Define the set of scaling factors as $\Delta \equiv \{\beta \in [0, 1]^n : \sum_{i \in N} \beta_i = 1\}$.

Theorem 1A. Assume the IVV assumption holds. With

$$\overline{W}(\hat{s}; \beta) = \sum_{i \in N} \int_{\hat{s}_i}^1 (\beta_i \phi_i(s) + w_i(s)) p(s) ds \quad (15)$$

the saddle value

$$\max_{\hat{s} \in [0, 1]^n} \widehat{W}^{NP}(\hat{s}) \equiv \max_{\hat{s} \in [0, 1]^n} \min_{\beta \in \Delta} \overline{W}(\hat{s}; \beta) = \min_{\beta \in \Delta} \max_{\hat{s} \in [0, 1]^n} \overline{W}(\hat{s}; \beta) \quad (16)$$

is the supremum of the principal's objective over all NIC mechanisms satisfying the **narrow participation constraints** and **limited liability at zero**.

If a maximizer of (16) does not have **coincident interior thresholds at distinct utilities**, the supremum is attained. Otherwise it is approached: for any maximizer $\hat{s}^*$ attaining $\max_{\hat{s} \in [0,1]^n} \widehat{W}^{NP}(\hat{s})$, there is a sequence of NIC-implementable threshold strategies $\hat{s}^k \rightarrow \hat{s}^*$ such that $\min_{\beta \in \Delta} \overline{W}(\hat{s}^k; \beta) \rightarrow \max_{\hat{s} \in [0,1]^n} \widehat{W}^{NP}(\hat{s})$.

Proof. In Appendix □

We modify the proof to drop the narrow participation constraints, retaining only limited liability at zero: $t(0, \dots, 0) \geq 0$. Given a threshold strategy $\hat{s}$, denote the set of the lowest threshold dimensions by $\underline{N}(\hat{s}) = \arg \min_{i \in N} \hat{s}_i$. Note that under a NIC-implementable threshold strategy and given there is an atom of probability on types that always choose zero, the narrow participation constraints holding implies that the limited liability at zero constraint holds also.¹⁰ Thus Theorem 1A is unchanged if we drop the limited liability at zero constraint. The atom of $s = 0$ types ensures that the principal's problem under narrow inference with the limited liability at zero constraint is not vacuous: without it they would be able to induce that no type chooses $(0, \dots, 0)$ and extract an arbitrarily large transfer payment from the other actions. The atom is irrelevant for Theorem 1A since the narrow participation constraints by themselves prevent this from occurring.

Theorem 1B. The value

$$\max_{\hat{s} \in [0,1]^n} \widehat{W}^{LL}(\hat{s}) = \max_{\hat{s} \in [0,1]^n} \left\{ \max_{j \in \underline{N}(\hat{s})} \int_{\hat{s}_j}^1 \phi_j(s) p(s) ds + \sum_{i \in N} \int_{\hat{s}_i}^1 w_i(s) p(s) ds \right\} \quad (17)$$

is the supremum of the principal's objective over all NIC mechanisms satisfying **limited liability at zero**.

If a maximizer has no **coincident thresholds below one at distinct utilities**, the supremum is attained. Otherwise it is approached: for any maximizer $\hat{s}^*$ of $\widehat{W}^{LL}(\hat{s})$, there is a sequence of NIC-implementable threshold strategies $\hat{s}^k \rightarrow \hat{s}^*$ such that $\widehat{W}^{LL}(\hat{s}^k) \rightarrow \widehat{W}^{LL}(\hat{s}^*)$.

Proof. In Appendix □

The following corollary gives a simple condition under which the supremum in Theorem 1A is always attained. We will see in Proposition 4 that the supremum in Theorem 1B is generally not attained. Under the limited liability at zero constraint only, there is a wide range of parameters where attaining the supremum would require inducing coincident thresholds with distinct utilities.

¹⁰This is explained in more detail in the proof of Theorems 1A and 1B.

Corollary 1. *Assume that $w_i(s) \neq 0$ for all $i \in N$ and $s \in (0, 1]$, and that IVV holds. Then we have that all $\hat{s}^{NP} \in \arg \max_{\hat{s} \in [0, 1]^n} \widehat{W}^{NP}(\hat{s})$ have no coincident interior thresholds at distinct utilities, and thus the supremum in Theorem 1A can be attained in the full problem.*

Proof. Let $\beta^* \in \Delta$ be the saddle point value from Theorem 1A. We have that $\hat{s}^{NP} \in \arg \max_{\hat{s} \in [0, 1]^n} \overline{W}(\hat{s}; \beta^*)$. Since $w_i(\cdot)$ is weakly increasing the assumption it does not cross zero means $w_i(s) > 0$ or $w_i(s) < 0$ for all $s \in (0, 1]$. For any dimension $i \in N$ with $\beta_i^* = 0$, it follows that $\hat{s}_i^{NP} \in \{0, 1\}$. For any dimensions $j \neq k$, $j, k \in N$ with $\beta_k^* > 0$, $\beta_j^* > 0$ we must have $v_k(\hat{s}_k^{NP})(1 - P(\hat{s}_k^{NP})) = v_j(\hat{s}_j^{NP})(1 - P(\hat{s}_j^{NP}))$ which implies that $v_k(\hat{s}_k^{NP}) = v_j(\hat{s}_j^{NP})$ if $\hat{s}_k^{NP} = \hat{s}_j^{NP} \in [0, 1)$. Thus any interior thresholds with coincident threshold must have coincident utilities. $\square$

To see some intuition for these results, consider the case with two dimensions and symmetric predictable utility $v_i(s) = v(s)$ and principal's direct utility $w_i(s) = w(s) > 0$ from actions across all dimensions $i \in N$. Here there are trivially no coincident interior thresholds at distinct utilities for every threshold strategy. With these utilities, it is optimal to set equal thresholds across dimensions in both the rational benchmark and under narrow inference for both the narrow participation and limited liability at zero constraint.¹¹

With a rational agent, from Proposition 1 the principal's optimum sets a threshold $\hat{s}$ so that the action is taken on all dimensions for all types above and the zero action is taken on all dimensions for all types below. This optimum is induced with a transfer mapping that is additive and identical across dimensions $t(a_1, a_2) = t^1(a_1) + t^2(a_2)$ and $t^1(1) = t^2(1) = \tilde{t}(1)$, $t^1(0) = t^2(0) = 0$. With a narrow agent, under the same strategy and transfer mapping the agents double count the effect of each action on the transfer. In each dimension, they believe that the transfer resulting from taking the action $a_i = 1$ is $2 \cdot \tilde{t}(1)$ and the transfer resulting from $a_i = 0$ is 0. This double-counting is the result of confounding neglect; the agent fails to adjust for the fact that every type who takes action $a_1 = 1$ also takes action $a_2 = 1$. The principal then has to halve the size of the difference in transfers in order to maintain the same thresholds $\frac{1}{2}\tilde{t}(1)$. This has the same effect as scaling the predictable utilities down by $\frac{1}{2}$ in each dimension when agents are rational. From Section 2.5, we see that this double counting effect on the incentive constraints would not emerge if the action distribution the agents are using to form beliefs were independent across dimensions.

¹¹When $w > 0$, this is demonstrated in Proposition 4 for the limited liability at zero case, and it is clear from the objective functions in the rational and narrow inference with narrow participation cases.

The fact that the type distribution positively correlates actions across dimensions is thus vital to the results.

The results extend this logic to asymmetric cases. With asymmetry, the scaling factors now differ across dimensions. The different participation regimes alter the form the factors take. Since narrow participation must simultaneously hold on every dimension, the tightest constraint across the dimensions caps the principal's objective value. The minimization of the objective value through the scaling factors reflects this cap. Limited liability at zero instead constrains only the agents who always take the zero actions. This means a constraint has to bind on only one dimension, and the principal can pick the most favourable one.

4.2 Preliminaries for Characterization Result

It will be useful in characterizing the principal's optimal mechanism under NIC to show how beliefs and the transfer mapping relate given any fixed distribution over actions. The following result shows when we can write the transfer distribution in terms of the beliefs over the expected transfer in either dimension. It gives a standard statistical correctness result that applies to beliefs formed using Bayesian Networks under perfect Directed Acyclic Graphs (DAGs) (Spiegler, 2020).

Lemma 1 (Statistical Correctness). *Given any distribution over actions g , for any two dimensions $i, j \in N$ we have that beliefs $\bar{t}_i, \bar{t}_j$ satisfy*

$$\sum_{a_i \in A_i} g_i(a_i) \bar{t}_i(a_i) = \sum_{a_j \in A_j} g_j(a_j) \bar{t}_j(a_j) \quad (18)$$

Proof. For any $i \in N$:

$$\sum_{a \in A} g(a) t(a) = \sum_{a_i \in A_i} g_i(a_i) \sum_{a_{-i} \in A_{-i}} \frac{g(a_i, a_{-i})}{g_i(a_i)} t(a_i, a_{-i}) = \sum_{a_i \in A_i} g_i(a_i) \bar{t}_i(a_i)$$

□

Given this, we call $\sum_{a \in A} g(a) t(a)$ the *mean transfer* from now on. For a fixed strategy and action distribution, this statistical correctness is necessary but not sufficient for a transfer mapping to exist that induces given beliefs. For an example of beliefs that satisfy the statistical correctness constraint but cannot be induced, consider the case with $N = \{1, 2\}$ and $g(1, 1) = g(0, 0) = \frac{1}{2}$. If beliefs do not also satisfy $\bar{t}_1(1) = \bar{t}_2(1)$ and $\bar{t}_1(0) = \bar{t}_2(0)$, then there is no transfer mapping implementing these beliefs under this action distribution.

The principal's objective can be written in terms of beliefs. This means it is useful to work directly with beliefs rather than the underlying transfer mapping when we characterize the principal's optimal mechanism. Although the statistical correctness constraint is not sufficient, it will be sufficient if the strategy of agents takes a threshold form and has no coincident interior thresholds at distinct utilities. I now show that any NIC strategy must take a threshold form. This differs from the rational case where a threshold strategy is optimal for the principal under the IVV assumption, but is not an implication of IC.

Lemma 2. *Every NIC strategy takes a **threshold form** for almost all $s \in S$.*

Proof. Let $\hat{U}_i(s, a_i) = v_i(s)a_i + \bar{t}_i(a_i)$. Clearly $\hat{U}_i(s, 1) - \hat{U}_i(s, 0) = v_i(s) + \bar{t}_i(1) - \bar{t}_i(0)$ is increasing in $s \in (0, 1]$. There is a threshold $\hat{s}_i$ such that $\hat{U}_i(s, 1) \geq \hat{U}_i(s, 0)$ if and only if $s \geq \hat{s}_i$, in which case type s will choose $a_i = 1$. $\square$

The next result shows that given a threshold strategy with no coincident interior thresholds at distinct utilities, a transfer mapping exists that is consistent with this strategy and beliefs that both satisfy statistical correctness and ensures the strategy is NIC.

Proposition 2. *Fix a threshold strategy g with thresholds $\hat{s}$ and denote $N_0(\hat{s}) \equiv \{j \in N : \hat{s}_j = 0\}$.*

1. *The threshold strategy g is NIC under beliefs $\bar{t}$ if and only if $\bar{t}_i(1) = \bar{t}_i(0) - v_i(\hat{s}_i)$ if $\hat{s}_i \in (0, 1)$, $\bar{t}_i(1) \geq \bar{t}_i(0) - v_i(0)$ if $\hat{s}_i = 0$ and $\bar{t}_i(1) \leq \bar{t}_i(0) - v_i(1)$ if $\hat{s}_i = 1$.*
2. *If in addition beliefs $\bar{t}$ are induced by transfer t with mean transfer T , then $\bar{t}_i(0) = T + V_i(\hat{s}_i)$ if $\hat{s}_i \in (0, 1)$, $\bar{t}_i(0) \leq T + V_i(\hat{s}_i)$ if $\hat{s}_i = 0$ and $\bar{t}_i(0) = T$ if $\hat{s}_i = 1$.*
3. *There exists a transfer mapping t under which g is NIC if and only if there are no coincident interior thresholds at distinct utilities: $v_i(\hat{s}_i) = v_j(\hat{s}_j)$ for any $i \neq j, i, j \in N$ such that $\hat{s}_i = \hat{s}_j \in (0, 1)$.*
4. *Suppose g has no coincident interior thresholds at distinct utilities. For every $T \in \mathbb{R}$ and every $c_0 \leq T + \min_{j \in N_0(\hat{s})} V_j(0)$ ¹² there is a transfer mapping t*

¹²If $N_0(\hat{s}) = \emptyset$ this restriction is vacuous as it only applies at zero threshold dimensions.

under which g is NIC and which induces the beliefs

$$\begin{aligned}\bar{t}_i(0) &= T + V_i(\hat{s}_i), & \bar{t}_i(1) &= T - P(\hat{s}_i)v_i(\hat{s}_i) & \text{if } \hat{s}_i \in (0, 1) \\ \bar{t}_j(0) &= c_0, & \bar{t}_j(1) &= \frac{T - P(0)c_0}{1 - P(0)} & \text{if } j \in N_0(\hat{s}) \\ \bar{t}_i(0) &= T & & & \text{if } \hat{s}_i = 1\end{aligned}$$

The restriction of t to $A(g) \equiv \{a \in A : \exists s \in S \text{ such that } g(a|s) > 0\}$ is uniquely determined by these beliefs, and t can be chosen additive across dimensions: $t(a) = \sum_{i \in N} t^i(a_i)$.

Proof. In Appendix □

The principal can exploit two features of the narrow agents' misperception: that they can only perceive of the transfer mapping as additive and that their beliefs do not account for confounding bias. Proposition 2 means that if the principal implements a threshold strategy, they have no payoff gain from implementing a transfer mapping that is not additive. Thus, the principal does not exploit the agents' potentially false perception of the transfer mapping as additive in their choice of optimal mechanism.

The transfer mapping can be constructed as additive because under a threshold strategy, there are action combinations that are not chosen by any type of agent. Take the case with two dimensions of action, and assume that the agents' strategy is such that the threshold for the first dimension is lower than the threshold in the second dimension; $\hat{s}_1 < \hat{s}_2$. Types in interval $[0, \hat{s}_1)$ choose actions $(0, 0)$, types in $[\hat{s}_1, \hat{s}_2)$ choose $(1, 0)$ and types in $[\hat{s}_2, 1]$ choose $(1, 1)$. The transfer for actions $(0, 1)$, chosen by no type, can be set so that the transfer mapping is additive. The case where $\hat{s}_1 > \hat{s}_2$ is symmetric. Proposition 2 extends this logic to the general case with any number of dimensions.

The proof of Theorems 1A and 1B works as follows. First, we show that due to the statistical correctness constraints the value functions in the theorem statements are upper bounds on the principal's objective in the full problem, and use Proposition 2 to show that the value functions can be attained as the principal's objective function in the full problem at threshold strategies satisfying the right condition on coincident thresholds. We then show that under narrow participation the value function takes a max-min form. Under IVV, we can apply a standard minimax theorem argument to obtain a saddle point, allowing us to interchange the min and the max. Finally, we show the value functions can always be approached by a sequence of NIC-implementable strategies.

4.3 Effect of Narrow Agents on the Principal's Welfare

Using the characterization of the principal's optimal mechanism, I show that when actions have predictable costs for all types of the agent then the principal is better off under narrow inference. Conversely, when actions have predictable benefits for all types then the principal is worse off.¹³

Proposition 3. *Assume the IVV assumption holds. Under both the narrow participation and limited liability at zero constraints*

1. *If for every $i \in N$ we have $v_i(s) < 0$ for all $s \in (0, 1]$, the principal can obtain at least as high an objective value when the agents are narrow compared to the rational benchmark.*
2. *If for every $i \in N$ we have $v_i(s) > 0$ for all $s \in (0, 1]$, the principal obtains at least as high an objective value in the rational benchmark compared to when the agents are narrow.*

Proof. [In Appendix](#) □

The intuition for this result is the same as that given for the scaling factors at the start of Section 4. Under narrow inference the agents' perception of the effect of their actions on the transfer is amplified due to their neglect of confounding. When the agents' actions are costly for the agent, their overestimation of the transfer under narrow inference means less transfer has to be given to higher types in order to implement any given strategy. When actions benefit the agent then the principal is a net recipient of transfers from the agent and the logic is reversed.¹⁴

Suppose that whether the agents do narrow or rational inference is something that the principal can influence or design. This could be either through the way they present the mechanism or data about the mechanism. If the default is that the agents do narrow inference, the principal can try to educate the agents on how to do rational inference. Proposition 3 then shows in what cases the principal would gain from de-biasing the agents and in what cases they would not. This manipulation of the choice architecture could be seen as a form of 'bundling' in the way choices are presented to the agent. In contrast to this soft presentational bundling, in Section

¹³In Section A.7 in the Online Appendix, we also study the effect of narrow inference on the average welfare of agents. The effect is generally ambiguous, and can be either positive or negative under all the conditions in this section.

¹⁴In Section A.6 of the Online Appendix, Proposition OA.1 shows that a version of Proposition 3 holds when we don't have IVV.

4.5 I explore when the principal can benefit from pure bundling — that is explicitly restricting the ability of agents to take certain actions separately.

4.4 Effect of Narrow Agents on the Implemented Strategy

The characterization results also allow us to see how the principal's optimal thresholds change when we move to narrow agents from the rational benchmark.

Proposition 4. *Assume the IVV assumption holds and that $w_i(s) \neq 0$ for all $i \in N$ and $s \in (0, 1]$. Let $\hat{s}^{NP}$ be a maximizer of the objective under narrow participation; $\hat{s}^{NP} \in \arg \max_{\hat{s} \in [0,1]^n} \widehat{W}^{NP}(\hat{s})$, $\hat{s}^{LL}$ be a maximizer of the objective under limited liability at zero; $\hat{s}^{LL} \in \arg \max_{\hat{s} \in [0,1]^n} \widehat{W}^{LL}(\hat{s})$ and $\hat{s}^{rat}$ be a maximizer of the objective with rational agents: $\hat{s}^{rat} \in \arg \max_{\hat{s} \in [0,1]^n} W(\hat{s})$. Then*

1. *For every $i \in N$ such that $w_i(s) > 0$ for all $s \in (0, 1]$, $\hat{s}_i^{NP} \leq \hat{s}_i^{rat}$, so $a_i = 1$ is taken by a weakly larger proportion of types.*
2. *For every $i \in N$ such that $w_i(s) < 0$ for all $s \in (0, 1]$, $\hat{s}_i^{NP} \geq \hat{s}_i^{rat}$, so $a_i = 1$ is taken by a weakly smaller proportion of types.*
3. *If for every $i \in N$ we have $w_i(s) > 0$ for all $s \in (0, 1]$, there is a common threshold such that $\hat{s}^c = \hat{s}_i^{LL}$ for all $i \in N$, and $\hat{s}^c \leq \max_{i \in N} \hat{s}_i^{rat}$.*
4. *If for every $i \in N$ we have $w_i(s) < 0$ for all $s \in (0, 1]$, then for one dimension j^* ; $\hat{s}_{j^*}^{LL} = \hat{s}_{j^*}^{rat}$, while for all other dimensions $i \in N \setminus \{j^*\}$ we have $\hat{s}_i^{LL} = 1$. Consequently, on every dimension $a_i = 1$ is taken by a weakly smaller proportion of types.*

Proof. In Appendix □

Remark. *The threshold profiles $\hat{s}^{LL}$ may not have coincident thresholds below one at distinct utilities.¹⁵ However, the comparative statics on how the thresholds under narrow and rational agents compare continue to hold for threshold strategies that approach $\hat{s}^{LL}$ and thus approach the supremum given in Theorem 1B.*

Under narrow inference with narrow participation, if actions have predictable benefits to the principal for all types, the principal implements a strategy with a lower type threshold for taking the action on any dimension than in the rational

¹⁵Since the assumptions of Corollary 1 hold, $\hat{s}^{NP}$ has no coincident interior thresholds at distinct utilities so is always NIC-implementable and attains the value in Theorem 1A.

benchmark. The opposite holds when actions have predictable costs to the principal for all types of agent.

In the first case narrow inference reduces the marginal cost to the principal of implementing that any given proportion of agents take the action on any dimension. Given the benefits of the actions to the principal, this lower marginal cost means the principal wants a higher proportion of agents to take the action. In the second case, where actions predictably cost the principal the principal has a lower marginal benefit — in terms of transfer received — from a higher proportion of agents taking the action, but the same cost. The principal then wants to reduce the proportion of agents taking the actions on all dimensions.

Under the limited liability at zero constraint only, we no longer have a saddle point characterization as in Theorem 1A, and thus no longer have fixed weights for each dimension which give the comparative statics in Part 1 and 2 of Proposition 4. However, we can use the characterization in Theorem 1B to obtain a similar result. If all action dimensions directly benefit the principal the threshold on all dimensions can be set to (approximately, given the constraint of NIC-implementability) the same level without reducing the expected transfer the principal gives or receives from the mechanism. This equalized threshold cannot exceed the maximum of the rational thresholds, otherwise the principal would attain a higher objective value setting this rational maximum threshold as the common threshold.¹⁶

When all action dimensions are directly costly to the principal, the principal can always improve their objective function from any threshold profile $\hat{s}$ by setting the threshold equal to one for all but one dimension. For the remaining dimension, the objective coincides with the case with rational agents. It is thus optimal to set this threshold equal to the same level as in the rational benchmark. Consequently, the result that narrow thresholds exceed the rational thresholds continues to hold under the limited liability at zero constraint only.

4.5 Pure Bundling Under Narrow Inference

I now explore whether the principal can benefit from linking or bundling the different choices of the agents together. Pure bundling means that the principal can ensure that if an agent takes a given action then they also have to take some subset of actions

¹⁶Note that if all actions directly benefit the principal then — unlike in the case with narrow participation constraints — it is possible to construct an example where the principal prefers to set higher thresholds when facing narrow agents. Such an example can be found in the Online Appendix, Section A.2.

on other dimensions. I assume the agents perceive linked choices as joint, that is if the principal restricts the agents to having to choose any set of actions together the agents do not perform narrow inference for each dimension separately but instead treat the linked action as being one dimension.¹⁷

Under the rational benchmark, the ability of the principal to implement pure bundling through directly restricting the action choices available to the agents is irrelevant as they could achieve the same effect through penalizing the action choices through the transfer mapping. Under narrow inference the agents perceive the transfer function as being additive, so do not respond to interactions in the transfer function required to bundle action choices.¹⁸

Bundling has no benefit to the principal in the rational benchmark. The principal is always able to achieve any bundling through the transfer mapping rather than directly restricting actions, and from Proposition 1 under IVV the principal's optimal mechanism is additively separable. In contrast, we show that under narrow inference pure bundling can help mitigate losses to the principal caused by narrow inference.

Formally we model pure bundling through a reduction of the space of dimensions into a merged one. We say that a dimension space M is a *merge* of $N \supset M$ if there is a partition of N according to the dimensions in M where for each $i \in M$, there is a partition cell $\mathcal{P}(i)$ so that $v_i(s) = \sum_{j \in \mathcal{P}(i)} v_j(s)$ and $w_i(s) = \sum_{j \in \mathcal{P}(i)} w_j(s)$. Thus any partition cell $\mathcal{P}(i)$ represents a bundle of actions from the full dimension space that the agent must take jointly if at all under pure bundling. Comparing the principal's best-case mechanism in the merged and full dimension spaces then allows us to see if they can benefit from pure bundling.

We modify the expressions for the values in Theorems 1A and 1B so they account for the dimension space in the problem.

Proposition 5. *Assume the IVV assumption holds and that the agents do narrow inference.*

1. *If for every $i \in N$ we have $w_i(s) > 0$ and $v_i(s) \leq 0$ for all $s \in (0, 1]$, then for any full dimension space N and merge $M \subset N$ we*

¹⁷Without this assumption the agent would be forced to take actions that they didn't realize they were choosing. For example if actions 1 and 2 were bundled and they chose action 1 as under narrow inference for that dimension alone, then they would be choosing action 2 but without the predictable utility from action 2 entering in their utility.

¹⁸For example, if the principal wants to ensure that action $a_1 = 1$ is never taken separately from $a_2 = 1$ through the transfer function then $t(a_1 = 1, a_2 = 0) = -\infty$ and $t(a_1 = 0, a_2 = 1) = -\infty$. In the rational benchmark this always deters agents from taking these action combinations but under narrow inference if there are some agents choosing $(a_1, a_2) = (1, 1)$ and none taking the actions separately then agents' perceive the transfer from each action as $\bar{t}_1(a_1) = \bar{t}_2(1) = t(1, 1) > -\infty$ and so may not be deterred from taking the actions separately in equilibrium.

have that $\max_{\hat{s} \in [0,1]^{|N|}} \widehat{W}^{NP}(\hat{s} : N) \geq \max_{\hat{s} \in [0,1]^{|M|}} \widehat{W}^{NP}(\hat{s} : M)$ and $\max_{\hat{s} \in [0,1]^{|N|}} \widehat{W}^{LL}(\hat{s} : N) \geq \max_{\hat{s} \in [0,1]^{|M|}} \widehat{W}^{LL}(\hat{s} : M)$.

2. If for every $i \in N$ we have $w_i(s) < 0$ and $v_i(s) \geq 0$ for all $s \in (0, 1]$, then for any full dimension space N and merge $M \subset N$ we have that $\max_{\hat{s} \in [0,1]^{|M|}} \widehat{W}^{NP}(\hat{s} : M) \geq \max_{\hat{s} \in [0,1]^{|N|}} \widehat{W}^{NP}(\hat{s} : N)$.

Proof. In [Appendix](#) □

Remark. Both parts compare suprema. Under narrow participation these are attained ([Corollary 1](#)). Under limited liability they need not be, and merging can render NIC-implementable a threshold strategy that was not implementable in the split space. The principal's attained payoff can therefore exceed the payoff attained in the split-space even in [Part 1](#), but by [Theorem 1B](#) only by an arbitrarily small amount.

Pure bundling reduces or eliminates the distortions in agents' perceptions of the transfer under narrow inference, and also restricts the set of strategies the principal can implement on the bundled dimensions. When the principal is a net contributor of transfers in equilibrium then neither of these channels is advantageous to them. However, in cases where the principal is a recipient of transfers in equilibrium, pure bundling can benefit the principal by increasing the average transfer they extract from the agents for any fixed thresholds. We can show that for any subset of bundled action dimensions, the principal increases the expected transfer they receive by preventing the types who did not take all of the actions from taking any of the actions in the subset. With the limited liability at zero constraint only, this is no longer true and having to implement the same threshold for the bundled dimensions can outweigh any gains.^{[19](#)}

5 Discussion

5.1 Narrow Inference and the Choice Architecture of Professional Service Firms

The analysis of this paper suggests that how narrow inference affects incentive design depends fundamentally on the direction in which the transfer flows between the

¹⁹In the online appendix [Section A.3](#), there are examples where under narrow inference and with only the limited liability at zero constraint, the effect on the principal's welfare of merging dimensions can go either way.

parties involved. A principal always prefers to unbundle and split actions which are costly to the agent but beneficial to the principal, but can sometimes benefit from bundling and merging actions which are beneficial to the agent but costly to the principal.

This fits a pattern we see at professional service firms, such as in law or consultancy. Professional service firms often have internal labour markets and allow their workers to increasingly select into different roles and skill specializations as their career progresses. For examples see the ‘free-market’ system used by some law firms (Westfahl and Wilkins, 2017, pp. 1721–1723) for internal staffing, and similar structures in consulting (McKinsey & Company, 2019; Boston Consulting Group, 2026). On the client facing side the logic of specialization is reversed, with internally separated specialized workers being presented externally as a coordinated product. Garicano and Hubbard (2009) find using US data that as market size increases, lawyers within firms become more field-specialized while the firms themselves do not. They suggest that the wide scope of these firms could be because clients value ‘one-stop shopping’, an explanation that seems consistent with the soft presentational bundling discussed in Section 4.3.

There is also evidence of more explicit ‘hard bundling’, as discussed in Section 4.5, in some service industries. Nanda and Pancak (2010) suggest that real estate brokers in the US pressure state governments to enact laws that require certain services be sold as part of a ‘minimum service’. We also see bundling when worker organizations, such as professional associations or trade unions, are able to exert control over the provision of services and effectively act as the principal. Occupational licensing restricts certain tasks as having to be performed together, and is often worker-initiated (Kleiner and Krueger, 2013).²⁰ Similarly, trade unions have at times fought for their members to have exclusive rights to perform certain bundles of tasks.²¹

Thus, we observe a pattern of unbundled and split organization of labour with bundling in the provision of services. Narrow Inference provides an explanation for this pattern that is based on agents’ limited ability to understand the effects of different choice dimensions on total compensation or prices. This can be complementary with an explanation based on existing theories of bundling, such as McAfee et al. (1989).

²⁰See The Economist (2018) for a discussion of the self-regulation of licensing by the legal profession in the United States. Also the American Medical Association detailing their campaigning to prevent ‘scope creep’ (Smith, 2025).

²¹See Cummins (1926) for an example of historical disputes involving the United Brotherhood of Carpenters and Joiners of America.

5.2 Model Assumptions and Possible Extensions

Separable Predictable Costs and Benefits

I assume the costs and benefits of the actions are additive and have no interactions across dimensions. This fits more cleanly with narrow inference. The agents perceiving complementarity or substitutability in the non-transfer part of their utility seems inconsistent with treating them as separate decision problems. A possible generalization would be to allow narrow inference misperceptions to extend to non-transfer components of the agents' payoff. Allowing interactions only in the principal's predictable utilities would also be consistent with narrow inference. However, this would mean the result from Proposition 1 that the principal's optimal mechanism takes a simple threshold form would not hold unless we make a more complicated assumption on the primitives than increasing virtual values (IVV).

Single Type Dimension

The assumption of a single dimension of type fits an interpretation that there is a single latent factor such as ability or company size affecting the agents' utilities. It is possible to express the primitives of the model so that each dimension has its own type variable as long as these variables are co-monotonically distributed.²² In general, allowing multiple type dimensions opens up the difficulties associated with solving multi-dimensional screening problems.²³ This can make the rational benchmark very complicated and thus prevents easy comparisons with the principal's choice of mechanism under narrow inference. However, I conjecture that versions of the results continue to hold if we restrict the principal to choosing an additive transfer mapping and the types are positively correlated across dimensions.

Homogeneity of Cognitive Type

The analysis of this paper is restricted to the cases where all types of agents are rational agents who fully understand the transfer mapping or all types of agents perform narrow inference. Analysis of the case with a mixture of both rational and boundedly rational agents is much more difficult as we lose the tractability that comes from being able to write the principal's objective and all the constraints in terms of the beliefs of the narrow agents only. With mixed cognitive types we would

²²Under this transformation the action $a_i = 1$ on any dimension has a type-independent predictable utility $q \in [v_i(0), v_i(1)]$ that is distributed according to cdf $\tilde{P}_i(q) = P(v_i^{-1}(q))$.

²³For examples, see (Rochet, 1987) or (Armstrong, 1996).

have to account for the beliefs of the rational and narrow agents about the transfer mapping when solving the problem, and the relationship between them would become one of the constraints. The dependence of this relationship (8) on the equilibrium distribution over actions g makes it complicated to solve for the principal's optimal mechanism under this nonlinear constraint. As such I leave studying narrow inference with mixed cognitive types to future work.

A Appendix

Proof of Proposition 1

As stated, this result adapts Proposition 3.1 of [Carroll \(2017\)](#) to the model of this paper. The differences between the two settings are described in Online Appendix Section [A.5](#).

Step 1 (envelope formula). For any $s, s' \in (0, 1]$, the incentive constraints (5) imply

$$U(s) \geq U(s') + \sum_{i \in N} (v_i(s) - v_i(s')) \sum_{a_i \in A_i} g_i(a_i | s') a_i$$

We can use these rewritten ICs to show an envelope condition holds by using the Lipschitz continuity arguments in Theorems 1 and 2 of [Milgrom and Segal \(2002\)](#).

$$U(s) = U(s') + \sum_{i \in N} \int_{s'}^s v'_i(z) \sum_{a_i \in A_i} a_i g_i(a_i | z) dz \quad (19)$$

in particular $U(s)$ is weakly increasing in $s \in (0, 1]$.

Step 2 (virtual-surplus form of the objective). Write the expected transfer as

$$\int_0^1 \sum_{a \in A} t(a) g(a | s) dP(s) = \int_0^1 U(s) dP(s) - \sum_{i \in N} \int_0^1 \sum_{a_i \in A_i} v_i(s) a_i g_i(a_i | s) dP(s)$$

Substituting this into (3) and switching the order of integration (the inactive types contributing nothing, as they take the zero action on every dimension), the principal's objective (3) becomes

$$W(t, g) = \sum_{i \in N} \int_0^1 \sum_{a_i \in A_i} (\phi_i(s) + w_i(s)) a_i g_i(a_i | s) dP(s) - U(0) \quad (20)$$

Step 3 (relaxed problem). Since $U(0) = t(0, \dots, 0)$, limited liability at zero is equivalent to $U(0) \geq 0$, and it is optimal to set $U(0) = 0$. Relaxing the problem by dropping IC and the requirement that the expected utilities be achievable given g , (20) is maximized pointwise over $s \in (0, 1]$ by $g_i(1 | s) = \mathbb{1}\{\phi_i(s) + w_i(s) \geq 0\}$. By IVV and the assumption that w_i is weakly increasing, $\phi_i(s) + w_i(s)$ is strictly increasing in s , so the pointwise maximizer is a threshold strategy $g_i(1 | s) = \mathbb{1}\{s \geq \hat{s}_i\}$ for some $\hat{s} \in [0, 1]^n$.

Step 4 (implementation). It remains to show the relaxed value is attained. Under

the transfer mapping in the proposition statement, the expected utility of type s from following the threshold strategy is

$$U(s) = \sum_{i \in N} (v_i(s) - v_i(\hat{s}_i)) \mathbb{1}\{s \geq \hat{s}_i\}$$

and deviating on any dimension i changes utility by $\pm(v_i(s) - v_i(\hat{s}_i))$, which is unprofitable on each side of the threshold since v_i is increasing. Hence the threshold strategy is IC, with $U(0) = t(0, \dots, 0) = 0$, so limited liability at zero holds and the relaxed bound is attained. The solutions of the relaxed and full problems therefore coincide.

Proof of Proposition 2

Part 1. Necessity holds by Lemma 2. To see sufficiency, let $\hat{U}_i(s, a_i) = v_i(s)a_i + \bar{t}_i(a_i)$ and $\hat{U}_i(s, 1) - \hat{U}_i(s, 0) = v_i(s) + \bar{t}_i(1) - \bar{t}_i(0)$. Given the threshold strategy, the beliefs in the proposition statement ensure that agents are indifferent between taking either action at any interior threshold, always weakly prefer 1 if $\hat{s}_i = 0$ and always weakly prefer 0 if $\hat{s}_i = 1$.

Part 2. By Lemma 1, statistical correctness gives $P(\hat{s}_i)\bar{t}_i(0) + (1 - P(\hat{s}_i))\bar{t}_i(1) = T$ for every dimension $i \in N$. Combining with the conditions in Part 1 gives the result. At $\hat{s}_i = 1$ the mean condition alone gives $\bar{t}_i(0) = T$. At $\hat{s}_i = 0$ the NIC inequality $\bar{t}_i(1) - \bar{t}_i(0) \geq -v_i(0)$ which combined with the mean condition rearranges to $\bar{t}_i(0) \leq T + V_i(0)$.

Preliminaries for the rest of the proof. Given the threshold strategy g , there are $0 \leq K \leq |N|$ distinct thresholds below one, which we label $0 \leq \tau^1 < \dots < \tau^{K-1} < \tau^K < 1$ with the conventions that $P(\tau^0) = 0$ and $\tau^{K+1} = 1$. Denote the set of all action combinations that have positive probability under g by $A(g) = \{a \in A : \exists s \in S \text{ such that } g(a|s) > 0\}$. Define the strict product order $\succ$ on actions A such that $a \succ a'$ if and only if $a_i \geq a'_i$ for all $i \in N$ and $a_j > a'_j$ for at least one $j \in N$. The inactive types $s = 0$ always take $\tilde{a}^0 = (0, \dots, 0)$. If $\tau^1 = 0$, they alone do so while if $\tau^1 > 0$ all types in the interval $[0, \tau^1)$ do so.

Claim 1. (i) Each interval has mass $P(\tau^{k+1}) - P(\tau^k) > 0$ (The atom at $s = 0$ ensures this for $k = 0$). (ii) In each interval $[\tau^k, \tau^{k+1})$ ($[\tau^K, 1]$ for $k = K$) almost all strictly positive types $s > 0$ take a common action $\tilde{a}^k$. (iii) $A(g)$ is a chain $\tilde{a}^0 \prec \tilde{a}^1 \prec \dots \prec \tilde{a}^K$ with $\tilde{a}^0 = (0, \dots, 0)$. (iv) Each dimension i with $\hat{s}_i < 1$ switches from 0 to 1 exactly once along the chain, at the index $l(i)$ with $\hat{s}_i = \tau^{l(i)}$. (v) $\bar{t}_j(0) = t(0, \dots, 0)$ for every $j \in \underline{N}(\hat{s})$.

Proof of Claim 1. If $K = 0$ all types choose $a = 0$, and every $\bar{t}_i(1)$ is off-path. This means all parts of the claim hold trivially with $t(0, \dots, 0) = T$. Therefore assume $K \geq 1$. Part (i) holds from the definition of a cdf. The definition of a threshold strategy ensures Part (ii). For Part (iii), take any $m > l$. For every $i \in N$ such that $\tilde{a}_i^l = 1$, there is a threshold $\hat{s}_i$ where $\tau^l \geq \hat{s}_i$ and since $\tau^m > \tau^l$ we have $\tau^m \geq \hat{s}_i$ and therefore $\tilde{a}_i^m = 1$ also. There is at least one dimension $j \in N$ such that $\hat{s}_j = \tau^m$ and thus for $s \in [\tau^m, 1]$ we have $g_j(1|s) = 1$ while for $s \in [0, \tau^m)$ we have $g_j(1|s) = 0$, which means $\tilde{a}_j^m = 1$ but $\tilde{a}_j^l = 0$. This also gives Part (iv). The inactive types $s = 0$ always take $\tilde{a}^0 = (0, \dots, 0)$. If $\tau^1 = 0$, they alone do so while if $\tau^1 > 0$ all types in the interval $[0, \tau^1)$ do so. For $j \in \underline{N}(\hat{s})$, a type choosing $a_j = 0$ lies below every threshold, hence chooses $(0, \dots, 0)$. By Part (i) this event has positive mass, so $\bar{t}_j(0) = t(0, \dots, 0)$, giving Part (v).

By Claim 1, statistical correctness from Lemma 1 and the definition of beliefs in (8) we can then write the beliefs in terms of the transfer mapping t at actions $a \in A(g)$ and the type distribution P evaluated at the cut offs τ . For any dimension i with $\hat{s}_i < 1$ we have

$$\bar{t}_i(0) = \frac{\sum_{m=0}^{l(i)-1} (P(\tau^{m+1}) - P(\tau^m)) t(\tilde{a}^m)}{P(\tau^{l(i)})} \quad (21)$$

$$\bar{t}_i(1) = \frac{T - \sum_{m=0}^{l(i)-1} (P(\tau^{m+1}) - P(\tau^m)) t(\tilde{a}^m)}{1 - P(\tau^{l(i)})} = \frac{T - P(\tau^{l(i)}) \bar{t}_i(0)}{1 - P(\tau^{l(i)})} \quad (22)$$

When $\hat{s}_i = 1$, $a_i = 1$ is off-path and $\bar{t}_i(1)$ is a free value, and by Proposition 2 part 2 we have that $\bar{t}_i(0) = T$.

Part 3 (Necessity): For any two dimensions $i, j \in N$ with identical thresholds $\hat{s}_i = \hat{s}_j \in (0, 1)$ under g , the expressions for beliefs in terms of transfers are given by (21) and (22). Since the expressions are identical for both dimensions we have $\bar{t}_i(0) = \bar{t}_j(0)$ and $\bar{t}_i(1) = \bar{t}_j(1)$. By Proposition 2 part 1, any interior threshold on dimension $h \in N$ must satisfy $\bar{t}_h(1) - \bar{t}_h(0) = -v_h(\hat{s}_h)$, so identical beliefs across dimensions $i \neq j$ when $\hat{s}_i = \hat{s}_j \in (0, 1)$ means we must have $v_i(\hat{s}_i) = v_j(\hat{s}_j)$.

Part 3 (Sufficiency) and Part 4: Suppose g satisfies the no coincidence at interior threshold condition. Let $T \in \mathbb{R}$ and $c_0 \leq T + \min_{j \in N_0(\hat{s})} V_j(\hat{s}_j)$. For each $l = 1, \dots, K$ we can define the target beliefs

$$c_l \equiv \begin{cases} T + V_i(\hat{s}_i) \text{ for any } i \text{ with } \hat{s}_i = \tau^l & \text{if } \tau^l > 0 \\ c_0 & \text{if } \tau^l = 0 \text{ (i.e. } l = 1, \tau^1 = 0) \end{cases}$$

So that for all $i \in N$ with $\hat{s}_i < 1$, $\bar{t}_i(0) = c_{l(i)}$ and if $\tau^{l(i)} < 1$, $\bar{t}_i(1) = \frac{T - P(\tau^{l(i)})c_{l(i)}}{1 - P(\tau^{l(i)})}$. These are equal to the beliefs in the proposition statement. Now define

$$M_l \equiv P(\tau^l)c_l \text{ for all } 1 \leq l \leq K, \quad M_0 \equiv 0, \quad M_{K+1} \equiv T \quad (23)$$

which is well defined: any two dimensions with the common interior threshold $\tau^l > 0$ share $v_i(\hat{s}_i)$ by hypothesis, and hence share $V_i(\hat{s}_i)$.

We can then set the true transfer given to any action $\tilde{a}^l \in A(g)$ equal to

$$t(\tilde{a}^l) = \frac{M_{l+1} - M_l}{P(\tau^{l+1}) - P(\tau^l)} \quad (24)$$

Given g , this transfer mapping induces the target beliefs. Denote the beliefs induced by t and g by $(\bar{t}_i^*(0), \bar{t}_i^*(1))_{i \in N}$. Using our expressions for beliefs in terms of transfers (21) and (22) with this transfer mapping and g gives

$$\begin{aligned} \bar{t}_i^*(0) &= \frac{\sum_{m \leq l(i)-1} (P(\tau^{m+1}) - P(\tau^m))t(\tilde{a}^m)}{P(\tau^{l(i)})} = \frac{\sum_{m \leq l-1} (M_{m+1} - M_m)}{P(\tau^{l(i)})} \\ &= \frac{P(\tau^{l(i)})c_l}{P(\tau^{l(i)})} = c_l \end{aligned} \quad (25)$$

We also have $\sum_{k=0}^K (P(\tau^{k+1}) - P(\tau^k))t(\tilde{a}^k) = \sum_{k=0}^K (M_{k+1} - M_k) = M_{K+1} - M_0 = T$. This means this transfer mapping together with g satisfies statistical correctness. When $\hat{s}_i < 1$, since we have shown $\bar{t}_i^*(0) = c_l$, by (22)

$$\bar{t}_i^*(1) = \frac{T - P(\tau^{l(i)})\bar{t}_i^*(0)}{1 - P(\tau^{l(i)})} = \frac{T - P(\tau^{l(i)})c_{l(i)}}{1 - P(\tau^{l(i)})}$$

When $\hat{s}_i = 1$ we have that $\bar{t}_i^*(1)$ is off-path, so we can set it equal to any $\bar{t}_i(1) \leq T - v_i(1)$. By Proposition 2 Part 1, we have that under these beliefs g is NIC. Thus, under this transfer mapping the threshold strategy g is NIC and the target beliefs are induced.

Uniqueness on $A(g)$: From 24 we have that $(P(\tau^{l+1}) - P(\tau^l))t(\tilde{a}^l) = M_{l+1} - M_l$ for $0 \leq l \leq K$, so the total transfer given to types in interval $[\tau^l, \tau^{l+1})$ is pinned by the beliefs. This means $t(\tilde{a}^l)$ is pinned by beliefs as $(P(\tau^{l+1}) - P(\tau^l)) > 0$ by definition of Claim 1 (i).

Additivity: Let $N_l \equiv \{i \in N : \hat{s}_i = \tau^l\}$ be the set of dimensions switching from 0 to 1 between $\tilde{a}^{l-1}$ and $\tilde{a}^l$. By Claim 1 (iv) the N_l are disjoint since each dimension

switches once. Choose any $k \in N_l$ and set $t^k(1) - t^k(0) = t(\tilde{a}^l) - t(\tilde{a}^{l-1})$, while setting $t^i(1) = t^i(0)$ for all other dimensions $i \in N \setminus \{k\}$ (including those with $\hat{s}_i = 1$). Set base levels $t^i(0) = t(\tilde{a}^0)/n$. Then for every $0 \leq l \leq K$, define an additive function $\sum_i t^i(\tilde{a}_i^l) = t(\tilde{a}^0) + \sum_{m=1}^l (t(\tilde{a}^m) - t(\tilde{a}^{m-1})) = t(\tilde{a}^l)$. This additive function agrees with t on $a \in A(g)$ and extends it to all of $a \in A$.

Proof of Theorems 1A and 1B

From Lemma 2, we know that every NIC strategy is a threshold strategy. Therefore we optimize over profiles of thresholds $\hat{s} \in [0, 1]^n$. Under a threshold strategy, the narrow participation constraints imply the limited liability at zero constraint. If $\bar{t}_i(0) \geq 0$ for all $i \in N$, then for $j \in \underline{N}(\hat{s})$, we have that $\bar{t}_j(0) = t(0, \dots, 0)$. This is because the action $a = 0$ is always chosen by some types (due to the atom at $s = 0$) and the fact that an agent who takes action $a_j = 0$ on dimensions $j \in \underline{N}(\hat{s})$ takes action zero on every dimension. Thus when we consider the problem under the narrow participation and limited liability at zero constraints, we can treat it as if only the narrow participation constraints must hold.

Define the *maximal extraction* under each participation constraint $\bullet \in \{NP, LL\}$ (where NP is narrow participation and LL is limited liability at zero) by

$$b^{NP}(\hat{s}) \equiv \min_{i \in N} V_i(\hat{s}_i), \quad b^{LL}(\hat{s}) \equiv \max_{j \in \underline{N}(\hat{s})} V_j(\hat{s}_j) \quad (26)$$

Let $\Pi^{\bullet,*}$ be the supremum of the principal's objective over all NIC-implementable threshold strategies (so those for which there exist a true transfer mapping consistent with the NIC strategy and beliefs) that satisfy the participation constraints $\bullet$. For any $\hat{s} \in [0, 1]^n$ define the value function

$$\widehat{W}^{\bullet}(\hat{s}) \equiv b^{\bullet}(\hat{s}) + \sum_{i \in N} \int_{\hat{s}_i}^1 w_i(s) p(s) ds \quad (27)$$

The proof has three lemmas. In Lemma A.1, we show that every NIC-implementable threshold strategy $\hat{s}$ gives the principal an objective value of at most $\widehat{W}^{\bullet}(\hat{s})$, so $\Pi^{\bullet,*} \leq \max_{\hat{s} \in [0, 1]^n} \widehat{W}^{\bullet}(\hat{s})$. Lemma A.1 also shows that this value $\widehat{W}^{\bullet}(\hat{s})$ can be attained in the full problem by any threshold profile satisfying the no coincident threshold condition that is appropriate to the $\bullet$ constraint regime. Lemma A.2 shows we can rewrite $\widehat{W}^{NP}(\hat{s})$ in the minimax form in the theorem statement. Finally, Lemma A.3 shows that $\Pi^{\bullet,*} = \max_{\hat{s} \in [0, 1]^n} \widehat{W}^{\bullet}(\hat{s})$ for all $\hat{s}$ that are NIC-implementable, and that in the full problem we can always approximate the value $\max_{\hat{s} \in [0, 1]^n} \widehat{W}^{\bullet}(\hat{s})$

arbitrarily closely by choosing nearby threshold profiles that are NIC-implementable.

Lemma A.1. *The value function $\widehat{W}^\bullet(\hat{s})$ has the following properties.*

1. *Any NIC mechanism inducing thresholds $\hat{s}$ and satisfying participation constraints $\bullet$ has an objective that is at most $\widehat{W}^\bullet(\hat{s})$. Hence $\Pi^{\bullet,*} \leq \max_{\hat{s} \in [0,1]^n} \widehat{W}^\bullet(\hat{s}) \equiv \widehat{W}^{\bullet,*}$.*
2. *Let $\hat{s} \in [0,1]^n$ have **no coincident interior thresholds at distinct utilities**. Then there is a NIC mechanism implementing $\hat{s}$ under which the principal's objective value is $\widehat{W}^{NP}(\hat{s})$.*
3. *Let $\hat{s} \in [0,1]^n$ have **no coincident thresholds below one at distinct utilities**, then there is a NIC mechanism implementing $\hat{s}$ under which the principal's objective value is $\widehat{W}^{LL}(\hat{s})$.*

Proof. Part 1. Fix a mechanism t that NIC-implements a strategy g , and satisfies participation constraints $\bullet$. Let $T = \int_0^1 \sum_{a \in A} t(a)g(a|s)dP(s)$ be the mean transfer.

By Lemma 1, statistical correctness holds, and the mean transfer must have a common value across all dimensions: $P(\hat{s}_i)\bar{t}_i(0) + (1 - P(\hat{s}_i))\bar{t}_i(1) = T$ for all $i \in N$. We can then use this to rewrite the principal's objective

$$\begin{aligned} W(t, g) &= - \int_0^1 \sum_{a \in A} t(a)g(a|s)dP(s) + \sum_{j \in N} \int_0^1 w_j(s) \sum_{a_j \in A_j} a_j g_j(a_j|s)dP(s) \\ &= -T + \sum_{j \in N} \int_{\hat{s}_j}^1 w_j(s) p(s) ds \end{aligned}$$

By Proposition 2 part 2, it must be the case that $\bar{t}_i(0) \leq T + V_i(\hat{s}_i)$ for all $i \in N$. Together with the narrow participation constraints, which require that $\bar{t}_i(0) \geq 0$ for all $i \in N$, we therefore have that $T \in [-\min_{i \in N} V_i(\hat{s}_i), \infty)$.

Under the limited liability at zero constraint that $t(0, \dots, 0) \geq 0$, every $j \in \underline{N}(\hat{s})$ must have $\bar{t}_j(0) = t(0, \dots, 0) \geq 0$. This is because if $\min_{i \in N} \hat{s}_i > 0$, all types in $[0, \min_{i \in N} \hat{s}_i)$ choose zero on all dimensions (since these types are below any threshold), and if $\min_{i \in N} \hat{s}_i = 0$ the atom of types at $s = 0$ who choose $a = 0$ ensures $a_j = 0$ is chosen by a positive mass of types for all $j \in \underline{N}(\hat{s})$ (so the beliefs $\bar{t}_j(0)$ are not free). Thus $\max_{j \in \underline{N}(\hat{s})} \bar{t}_j(0) \geq 0$.²⁴ Combined with the fact that $\bar{t}_i(0) \leq T + V_i(\hat{s}_i)$ for all $i \in N$, this means $T \in [-\max_{j \in \underline{N}(\hat{s})} V_j(\hat{s}_j), \infty)$.

²⁴Note that this is an implication of limited liability at zero, but is not sufficient for it.

In both cases, the lower bound on T means that $W(t, g) \leq \widehat{W}^\bullet(\hat{s})$ for any NIC mechanism t implementing some threshold strategy $\hat{s}$ that satisfies participation constraint $\bullet$, which gives the result.

Part 2. By Proposition 2, we can define beliefs and a transfer mapping such that $\hat{s}$ is NIC-implementable and the common mean transfer is $T = -b^{NP}(\hat{s})$. Plugging this T into $W(t, g)$ gives the value $\widehat{W}^{NP}(\hat{s})$.

Part 3. Do the same exercise as in part 2, except with $T = -b^{LL}(\hat{s})$. The stronger condition is required here as if $\hat{s}_i = \hat{s}_j = 0$ for any $i \neq j, i, j \in N$ the threshold strategy can still be NIC-implementable under a valid transfer mapping by Proposition 2, but it requires a transfer mapping that does not attain the value given in the Theorem 1B statement. Strengthening to require no distinct utilities when $\hat{s}_i = \hat{s}_j = 0$ resolves this. $\square$

Lemma A.2. Let $\Delta = \{\beta \in [0, 1]^n : \sum_{i \in N} \beta_i = 1\}$ and $\overline{W}(\hat{s}; \beta) = \sum_{i \in N} \int_{\hat{s}_i}^1 (\beta_i \phi_i(s) + w_i(s)) p(s) ds$. We have that

$$\max_{\hat{s} \in [0, 1]^n} \widehat{W}^{NP}(\hat{s}) = \min_{\beta \in \Delta} \max_{\hat{s} \in [0, 1]^n} \overline{W}(\hat{s}; \beta) = \max_{\hat{s} \in [0, 1]^n} \min_{\beta \in \Delta} \overline{W}(\hat{s}; \beta) \quad (28)$$

Proof. Since $b^{NP}(\hat{s}) = \min_{i \in N} V_i(\hat{s}_i) = \min_{\beta \in \Delta} \sum_{i \in N} \beta_i V_i(\hat{s}_i)$, and since $\overline{W}(\hat{s}; \beta) = \sum_{i \in N} \beta_i V_i(\hat{s}_i) + \sum_{i \in N} \int_{\hat{s}_i}^1 w_i(s) p(s) ds$ it must hold that $\widehat{W}^{NP}(\hat{s}) = \min_{\beta \in \Delta} \overline{W}(\hat{s}; \beta)$. Thus, $\max_{\hat{s} \in [0, 1]^n} \widehat{W}^{NP}(\hat{s}) = \max_{\hat{s} \in [0, 1]^n} \min_{\beta \in \Delta} \overline{W}(\hat{s}; \beta)$.

Denote the strictly increasing inverse of P on $[P(0), 1]$ by P^{-1} . We can rewrite $\overline{W}$ in terms of the quantiles of the type distribution. Let $x_i = P(\hat{s}_i)$ and $x = (x_i)_{i \in N} \in [P(0), 1]^n$.

$$\overline{W}(P^{-1}(x); \beta) = \sum_{i \in N} \int_{x_i}^1 (\beta_i \phi_i(P^{-1}(u)) + w_i(P^{-1}(u))) du \quad (29)$$

Taking derivatives with respect to the threshold x_j gives

$$-(\beta_j \phi_j(P^{-1}(x_j)) + w_j(P^{-1}(x_j)))$$

By the IVV assumption, for fixed $\beta \in \Delta$ this is decreasing and thus $\int_{x_j}^1 (\beta_j \phi_j(P^{-1}(u)) + w_j(P^{-1}(u))) du$ is concave in $x \in [P(0), 1]^n$. It is also linear (hence convex) in β for fixed $x \in [P(0), 1]^n$. Thus, we can apply the minimax theorem (Sion, 1958) which gives the result. $\square$

Lemma A.3. For fixed $\bullet \in \{NP, LL\}$ let $\widehat{W}^{\bullet, *} \equiv \max_{\hat{s} \in [0, 1]^n} \widehat{W}^\bullet(\hat{s})$.

1. $\widehat{W}^{NP}(\hat{s})$ is continuous in $\hat{s} \in [0, 1]^n$ and $\widehat{W}^{LL}(\hat{s})$ is upper-semicontinuous in $[0, 1]^n$. Thus $\widehat{W}^{\bullet,*}$ is attained by some $\hat{s}^{\bullet,*} \in [0, 1]^n$.

2. Define

$$\begin{aligned}\mathcal{C}^{NP} &\equiv \{\hat{s} \in [0, 1]^n : i \neq j \in N \text{ with } \hat{s}_i = \hat{s}_j \in (0, 1) \Rightarrow v_i(\hat{s}_i) = v_j(\hat{s}_j)\} \\ \mathcal{C}^{LL} &\equiv \{\hat{s} \in [0, 1]^n : i \neq j \in N \text{ with } \hat{s}_i = \hat{s}_j \in [0, 1) \Rightarrow v_i(\hat{s}_i) = v_j(\hat{s}_j)\}\end{aligned}$$

For both $\bullet \in \{NP, LL\}$ the threshold strategies in $[0, 1]^n \setminus \mathcal{C}^\bullet$ lie in $\mathcal{D} \equiv \{\hat{s} \in [0, 1]^n : \hat{s}_i = \hat{s}_j \text{ for some } i \neq j \in N\}$, which is a closed and nowhere dense set. Hence $\mathcal{C}^\bullet \supseteq [0, 1]^n \setminus \mathcal{D}$ is dense in $[0, 1]^n$.

3. $\Pi^{\bullet,*} = \sup\{\widehat{W}^\bullet(\hat{s}) : \hat{s} \in \mathcal{C}^\bullet\} = \max_{\hat{s} \in [0, 1]^n} \widehat{W}^\bullet(\hat{s}) \equiv \widehat{W}^{\bullet,*}$, and for every $\epsilon > 0$ there is an implementable $\hat{s}$ with $\widehat{W}^\bullet(\hat{s}) + \epsilon \geq \Pi^{\bullet,*}$.

4. If a maximizer $\hat{s}^{\bullet,*}$ has no coincident (interior for $\bullet = NP$, below one for $\bullet = LL$) thresholds at distinct utilities, then $\hat{s}^{\bullet,*}$ is NIC-implementable and $\Pi^{\bullet,*}$ is attained. Moreover, for every maximizer there is a sequence of NIC-implementable strategies $\hat{s}^k \rightarrow \hat{s}^{\bullet,*}$ such that $\widehat{W}^\bullet(\hat{s}^k) \rightarrow \widehat{W}^{\bullet,*}$.

Proof. Part 1. In Theorem 1A, $\overline{W}(\hat{s}; \beta)$ is jointly continuous and $\beta \in [0, 1]^n, \sum_{i \in N} \beta_i = 1$ is a compact set, so $\widehat{W}^{NP}(\hat{s}) = \min_{\beta \in [0, 1]^n, \sum_{i \in N} \beta_i = 1} \overline{W}(\hat{s}; \beta)$ is continuous in $\hat{s}$. In Theorem 1B, $\widehat{W}^{LL}(\hat{s})$ is upper-semicontinuous: given any convergent sequence $\hat{s}^k \rightarrow \hat{s}$, we can find a subsequence on which a fixed j attains $\max_{l \in \underline{N}(\hat{s}^k)} V_l(\hat{s}_l^k)$; then $\hat{s}_j^k = \min_i \hat{s}_i^k \rightarrow \min_i \hat{s}_i$, so $j \in \underline{N}(\hat{s})$ and $\widehat{W}^{LL}(\hat{s}^k) = V_j(\hat{s}_j^k) + \sum_{i \in N} \int_{\hat{s}_i^k}^1 w_i(s) p(s) ds \rightarrow V_j(\hat{s}_j) + \sum_{i \in N} \int_{\hat{s}_i}^1 w_i(s) p(s) ds \leq \widehat{W}^{LL}(\hat{s})$. Thus both cases involve maximizing an upper-semicontinuous function over the compact set $[0, 1]^n$, which means the maximum is attained.

Part 2. The set $\mathcal{D}$ is contained in a finite union of hyperplanes, and so is closed with empty interior in $[0, 1]^n$. Thus its complement $[0, 1]^n \setminus \mathcal{D}$ is open and dense in $[0, 1]^n$. Thus we have that $\mathcal{C}^\bullet \supseteq [0, 1]^n \setminus \mathcal{D}$ is also dense in $[0, 1]^n$.

Parts 3 and 4. By Lemma A.1, for $\hat{s}$ satisfying the $\bullet$ required no coincident threshold condition the principal can attain value $\widehat{W}^\bullet(\hat{s})$. Hence $\Pi^{\bullet,*} \geq \sup\{\widehat{W}^\bullet(\hat{s}) : \hat{s} \in \mathcal{C}^\bullet\}$. We have from Lemma A.1 that $\Pi^{\bullet,*} \leq \widehat{W}^{\bullet,*}$.

By Lemma A.3 Part 1 there is always a maximizer of $\widehat{W}^\bullet(\hat{s})$ in $[0, 1]^n$ which we denote $\hat{s}^{\bullet,*}$. Take a sequence with distinct thresholds $(\hat{s}^k)_{k=0}^\infty$ that converges to $\hat{s}^{\bullet,*}$. For Theorem 1B we choose this sequence so that a given dimension $m \in$

$\arg \max_{j \in \underline{N}(\hat{s}^*)} V_j(\hat{s}_j^*)$ is always the lowest threshold for every $\hat{s}^k$ in the sequence. We know from Part 2 that every $\hat{s}^k$ in the sequence attains the value $\widehat{W}^\bullet(\hat{s}^k)$, and from the regularity of $\widehat{W}^\bullet(\hat{s})$ derived in Part 1 that $\widehat{W}^\bullet(\hat{s}^k) \rightarrow \widehat{W}^\bullet(\hat{s}^{\bullet,*}) = \widehat{W}^{\bullet,*}$. This means $\widehat{W}^{\bullet,*} = \sup\{\widehat{W}^\bullet(\hat{s}) : \hat{s} \in \mathcal{C}^\bullet\}$. This means $\Pi^{\bullet,*} = \widehat{W}^{\bullet,*}$, giving our results. $\square$

From these results, we have that Theorems 1A and 1B hold. For both participation constraints $\bullet \in \{NP, LL\}$ we have that the supremum of the principal's objective over all NIC-implementable mechanisms under participation constraints $\bullet$, denoted $\Pi^{\bullet,*}$, is equal to $\widehat{W}^{\bullet,*} \equiv \max_{\hat{s} \in [0,1]^n} \widehat{W}^\bullet(\hat{s})$, which is the value given in the Theorem statements. Moreover by Part 4 of Lemma A.3, we have that this supremum is attained exactly if a maximizer attaining $\widehat{W}^{NP,*}$ has no coincident interior thresholds at distinct utilities (and for $\widehat{W}^{LL,*}$ has no coincident thresholds below one at distinct utilities), and any maximizer is always approached by a sequence of NIC-implementable threshold strategies in any case.

Proof of Proposition 3

Fix a threshold strategy $\hat{s} \in [0,1]^n$. For both the narrow participation and limited liability at zero cases, there is some $\beta(\hat{s}) \in [0,1]^n$, $\sum_{i \in N} \beta(\hat{s})_i = 1$ such that we can write the principal's values from Theorems 1A and 1B as²⁵

$$W(\hat{s}; \beta(\hat{s})) = \sum_{i \in N} \int_{\hat{s}_i}^1 [\beta(\hat{s})_i \phi_i(s) + w_i(s)] p(s) ds$$

Take the difference between this and the principal's objective value in the rational case

$$\begin{aligned} & W(\hat{s}) - W(\hat{s}; \beta(\hat{s})) \\ &= \sum_{i \in N} \int_{\hat{s}_i}^1 [(\phi_i(s) + w_i(s)) - (\beta(\hat{s})_i \phi_i(s) + w_i(s))] p(s) ds \\ &= \sum_{i \in N} (1 - \beta(\hat{s})_i) \int_{\hat{s}_i}^1 (v_i(s) - \frac{1 - P(s)}{p(s)} v_i'(s)) p(s) ds \\ &= \sum_{i \in N} (1 - \beta(\hat{s})_i) v_i(\hat{s}_i) (1 - P(\hat{s}_i)) \end{aligned}$$

When $v_i(s) < 0$ clearly we have $W(\hat{s}) \leq W(\hat{s}; \beta(\hat{s}))$ for any $\beta(\hat{s})$ and for the sec-

²⁵In the narrow participation case this is the β^* from the saddle point in Theorem 1A, while in the limited liability at zero case this is any $\beta(\hat{s}) \in \{0,1\}^n$ satisfying $\beta(\hat{s})_j = 1$ for some $j \in \arg \max_{i \in \underline{N}(\hat{s})} V_i(\hat{s}_i)$ and $\beta(\hat{s})_k = 0$ for the remaining dimensions $k \in N \setminus \{j\}$.

ond case where $0 < v_i(s)$ we have $W(\hat{s}) \geq W(\hat{s}; \beta(\hat{s}))$ for any $\beta(\hat{s})$. Now, $W(\hat{s}; \beta(\hat{s}))$ may not actually be attained by any NIC-implementable thresholds. For Part 2, the result goes through as this means the actual attained objective value is less than $W(\hat{s}; \beta(\hat{s}))$ which is less than the objective value attained at $\hat{s} \in [0, 1]^n$ in the rational case.

For Part 1, if $\sum_{i \in N} (1 - \beta(\hat{s})_i) v_i(\hat{s}_i) (1 - P(\hat{s}_i)) < 0$ the result also goes through, as for any $\epsilon > 0$ we can attain $W(\hat{s}; \beta(\hat{s})) - \epsilon$ in the narrow case, and by choosing ϵ small we preserve $W(\hat{s}; \beta(\hat{s})) > W(\hat{s})$. When $\sum_{i \in N} (1 - \beta(\hat{s})_i) v_i(\hat{s}_i) (1 - P(\hat{s}_i)) = 0$, as $v_k(s) < 0$ for all $s \in (0, 1]$ and $k \in N$, we must have $\hat{s}_i = 1$ for any dimension i with $\beta(\hat{s})_i < 1$. This means at most one dimension j can have $\hat{s}_j \in [0, 1)$, since at most one dimension can have $\beta(\hat{s})_j = 1$ as $\sum_{i \in N} \beta(\hat{s})_i = 1$. Consequently, any threshold strategy at which the narrow and rational objective values are equal satisfies the no coincident interior thresholds at distinct utilities condition, and so is NIC-implementable. Thus, the principal is at least as well-off under narrow inference.

Proof of Proposition 4

Let β^* be the saddle-point shrinkage from Theorem 1A.

Part 1. If $\hat{s}_i^{rat} = 1$, the result holds trivially. We must have $\lim_{s \downarrow \hat{s}_i^{rat}} \{\phi_i(s) + w_i(s)\} \geq 0$ if $\hat{s}_i^{rat} \in [0, 1)$ with equality for an interior threshold. Thus $\phi_i(\hat{s}_i^{rat}) < 0$ if $\hat{s}_i^{rat} \in (0, 1)$ and therefore (by interior continuity) $0 \leq \phi_i(\hat{s}_i^{rat}) + w_i(\hat{s}_i^{rat}) \leq \beta_i^* \phi_i(\hat{s}_i^{rat}) + w_i(\hat{s}_i^{rat})$. If $\hat{s}_i^{rat} = 0$, we may have $\lim_{s \downarrow 0} \phi_i(s) \geq 0$, then $0 \leq \lim_{s \downarrow 0} \{\beta_i^* \phi_i(s) + w_i(s)\}$ as $\lim_{s \downarrow 0} w_i(s) \geq 0$ by assumption. Suppose that for some dimension $j \in N$, we have $\hat{s}_j^{NP} > \hat{s}_j^{rat}$. Since $\phi_j(s)$ is strictly increasing by IVV and $w_j(s)$ is also weakly increasing by assumption, we have that $\beta_j^* \phi_j(s) + w_j(s) > 0$ for all $s \in (\hat{s}_j^{rat}, \hat{s}_j^{NP}]$, a contradiction as $\beta_j^* \phi_j(\hat{s}_j^{NP}) + w_j(\hat{s}_j^{NP}) \leq 0$ for an optimal $\hat{s}_j^{NP} \in (0, 1]$.

Part 2. This holds via an argument symmetric to Part 1.

Part 3. As before let $\underline{N}(\hat{s}^{LL}) \equiv \arg \min_{i \in N} \hat{s}_i^{LL}$ and let $j^* \in \arg \max_{i \in \underline{N}(\hat{s}^{LL})} v_i(\hat{s}_i^{LL})(1 - P(\hat{s}_i^{LL}))$. At the threshold strategy that maximizes the value function, we cannot have any dimension $k \notin \underline{N}(\hat{s}^{LL})$. If $\hat{s}_k^{LL} > \hat{s}_{j^*}^{LL}$, then since $w_k(s) > 0$ for all $s \in (0, 1]$ we can reduce $\hat{s}_k^{LL}$ slightly and increase the value of $\widehat{W}^{LL}$, contradicting optimality.

Define the common threshold value $w^c(\hat{s}) \equiv \max_{j \in N} V_j(\hat{s}) + \sum_{i \in N} \int_{\hat{s}}^1 w_i(s) p(s) ds$. If we have a common threshold under the limited liability at zero constraint, it must maximize this value: $\hat{s}^c \in \arg \max_{\hat{s} \in [0, 1]} w^c(\hat{s})$. Let $j^* \in \arg \max_{j \in N} V_j(\hat{s}^c)$. Now suppose that $\hat{s}^c > \max_{i \in N} \hat{s}_i^{rat}$. Suppose we take

the objective function if the common threshold value were $\hat{s}_{max}^{rat} = \max_{i \in N} \hat{s}_i^{rat}$. We have that

$$\begin{aligned} w^c(\hat{s}_{max}^{rat}) &\equiv \max_{j \in N} \int_{\hat{s}_{max}^{rat}}^1 \phi_j(s) p(s) ds + \sum_{i \in N} \int_{\hat{s}_{max}^{rat}}^1 w_i(s) p(s) ds \\ &\geq \int_{\hat{s}_{max}^{rat}}^1 \phi_{j^*}(s) p(s) ds + \sum_{i \in N} \int_{\hat{s}_{max}^{rat}}^1 w_i(s) p(s) ds \end{aligned}$$

which means that, due to the definition of j^*

$$w^c(\hat{s}_{max}^{rat}) - w^c(\hat{s}^c) \geq \int_{\hat{s}_{max}^{rat}}^{\hat{s}^c} \phi_{j^*}(s) p(s) ds + \sum_{i \in N} \int_{\hat{s}_{max}^{rat}}^{\hat{s}^c} w_i(s) p(s) ds$$

From IVV, we have that $\phi_{j^*}(s) + w_{j^*}(s) > 0$ on $s \in [\hat{s}_{j^*}^{rat}, 1]$, which means it holds on $s \in [\hat{s}_{max}^{rat}, \hat{s}^c]$ also since $[\hat{s}_{max}^{rat}, \hat{s}^c] \subseteq [\hat{s}_{j^*}^{rat}, 1]$. Together with the fact $w_k(s) > 0$ for all $k \in N \setminus \{j^*\}$ this gives that $w^c(\hat{s}_{max}^{rat}) - w^c(\hat{s}^c) > 0$, which contradicts the definition of $\hat{s}^c$ as the maximizer of the objective. Consequently we must have that $\hat{s}^c \leq \max_{i \in N} \hat{s}_i^{rat} \equiv \hat{s}_{max}^{rat}$.

Part 4. Fix any threshold strategy $\hat{s}$. Again choose some $j^* \in \arg \max_{i \in \widehat{N}(\hat{s})} V_i(\hat{s}_i)$. Since $w_i(s) < 0$ for all $i \in N$, $s \in (0, 1]$ we can raise the value of $\widehat{W}^{LL}(\hat{s})$ by switching to a threshold strategy that sets $\hat{s}_k = 1$ for all $k \neq j^*$. Restricted to threshold strategies that have threshold equal to one for all but one dimension i , the objective function $\widehat{W}^{LL}$ then coincides with the rational objective function over this single dimension i . Thus it is maximized by setting $\hat{s}_i^{LL} = \hat{s}_i^{rat}$.

Proof of Proposition 5

For Parts 1 and 2, we now index the principal's value function from Theorem 1A by the dimensional space M : $\overline{W}(\hat{s}; \beta : M) = \sum_{i \in M} \int_{\hat{s}_i}^1 (\beta_i \phi_i(s) + w_i(s)) p(s) ds$. Similarly, define $\Delta^M = \{\beta \in [0, 1]^{|M|} : \sum_{i \in M} \beta_i = 1\}$.

Part 1. We show that for any fixed threshold strategy $(\hat{s}_i)_{i \in M}$ moving from any merged dimension space $M \subset N$ into the full dimension space N weakly increases the principal's objective value in either the narrow participation constraint case or the limited liability at zero case.

For the narrow participation constraint case, take a threshold strategy $\hat{s}^M = (\hat{s}_i)_{i \in M}$ in the merged space. We can increase the principal's value function from Theorem 1A by moving from merge M to the full dimension space N , whilst keeping

the same thresholds on the new split dimensions: $\hat{s}_i = \hat{s}_j$ for all $i \in N$ and $j \in \mathcal{P}(i)$.

$$\begin{aligned}
\min_{\beta \in \Delta^M} \overline{W}(\hat{s}; \beta : M) &\equiv \min_{\beta \in \Delta^M} \sum_{i \in M} \beta_i v_i(\hat{s}_i)(1 - P(\hat{s}_i)) + \sum_{i \in M} \int_{\hat{s}_i}^1 w_i(s) p(s) ds \\
&= \min_{\beta \in \Delta^M} \sum_{i \in M} \beta_i (1 - P(\hat{s}_i)) \sum_{j \in \mathcal{P}(i)} v_j(\hat{s}_j) + \sum_{i \in M} \int_{\hat{s}_i}^1 w_i(s) p(s) ds \\
&\leq \min_{\beta \in \Delta^N} \sum_{i \in M} (1 - P(\hat{s}_i)) \sum_{j \in \mathcal{P}(i)} \beta_j v_j(\hat{s}_j) + \sum_{i \in M} \int_{\hat{s}_i}^1 w_i(s) p(s) ds \\
&\equiv \min_{\beta \in \Delta^N} \overline{W}(\hat{s}; \beta : N)
\end{aligned}$$

Where the inequality holds as when $v_i(s) \leq 0$ for all $i \in N$, $s \in (0, 1]$, then for any $i \in M$ and $j \in \mathcal{P}(i)$, $0 \geq v_j(s) \geq v_i(s) = \sum_{k \in \mathcal{P}(i)} v_k(s)$ for all $s \in (0, 1]$.

For the limited liability at zero only case, when $w_i(s) > 0$ for all $s \in (0, 1]$ by part 3 of Proposition 4 we have that the threshold strategy that maximizes the value in 1B is equal across dimensions at some common value. The value function from Theorem 1B is equal to $w^c(\hat{s}^c : N) \equiv \max_{j \in N} V_j(\hat{s}^c) + \sum_{i \in N} \int_{\hat{s}^c}^1 w_i(s) p(s) ds$ at any common threshold $\hat{s}^c \in (0, 1]$ when the dimension space is N . For any merge M of the full dimension space N , we have for every $j \in N$ there is an $i \in M$ such that $j \in \mathcal{P}(i)$ and $0 \geq v_j(s) \geq v_i(s) = \sum_{k \in \mathcal{P}(i)} v_k(s)$ for all $s \in (0, 1]$. This means that $\max_{j \in N} V_j(\hat{s}^c) \geq \max_{j \in M} V_j(\hat{s}^c)$ and therefore that $w^c(\hat{s}^c : N) \geq w^c(\hat{s}^c : M)$ pointwise for any $\hat{s}^c$, which gives the result.

Part 2. The principal also faces a loss under the merged dimension space from a reduction in the space of threshold strategies that can be implemented. This is because under the merged space for any given dimension $i \in M$, all the thresholds $\hat{s}_j$, $j \in \mathcal{P}(i)$ must be equalized.

Given a fixed threshold strategy in the full dimension space $(\hat{s}_j)_{j \in N}$, define a threshold strategy in the merged space such that $\hat{s}_i = \max_{j \in \mathcal{P}(i)} \hat{s}_j$. For any $j(i) \in \arg \max_{k \in \mathcal{P}(i)} \hat{s}_k$ we can focus on solutions to the full dimension space minimax problem in Theorem 1A with $\beta_{j(i)}^* > 0$. If $\beta_{j(i)}^* = 0$ it would be optimal for the principal to implement the threshold $\hat{s}_{j(i)} = 1$ because $w_{j(i)}(s) < 0$ for all $s \in (0, 1]$. If $\hat{s}_j = 1$ for any dimension, then as $V_k(s) \geq 0$ for all $k \in N$ and $s \in [0, 1)$ at the solution to the minimax problem we can only have $\beta_h^* > 0$ for dimensions where $V_h(\hat{s}_h) = 0$. As $w_i(s) < 0$ for all $i \in N$, this means the principal's payoff is at most zero under the split space, which gives the result as the principal can obtain a zero payoff under the merged space.

With $\beta_{j(i)}^* > 0$ for every $i \in M$, we can show that the principal benefits from

moving from the threshold strategy $(\hat{s}_j)_{j \in N}$ in the full dimension space to the threshold strategy with $\hat{s}_i = \max_{j \in \mathcal{P}(i)} \hat{s}_j$ for all $i \in M$ in the merged space. This then gives the result.

$$\begin{aligned}
& \min_{\beta \in [0,1]^n : \sum_{i \in N} \beta_i = 1} \max_{(\hat{s}_j)_{j \in N} \in [0,1]^n} \overline{W}(\hat{s}; \beta : N) \\
&= \min_{k \in M} v_j(k)(\hat{s}_j^*(k))(1 - P(\hat{s}_j^*(k))) + \sum_{j \in N} \int_{\hat{s}_j^*}^1 w_j(s) p(s) ds \\
&\leq \min_{k \in M} \sum_{h \in \mathcal{P}(k)} v_h(\hat{s}_j^*(k))(1 - P(\hat{s}_j^*(k))) + \sum_{i \in M} \int_{\hat{s}_i^*}^1 \sum_{j \in \mathcal{P}(i)} w_j(s) p(s) ds \\
&= \min_{\beta \in [0,1]^m : \sum_{i \in M} \beta_i = 1} \sum_{i \in M} \beta_i v_i(\hat{s}_j(i))(1 - P(\hat{s}_j(i))) + \sum_{i \in M} \int_{\hat{s}_i^*}^1 w_i(s) p(s) ds \\
&\leq \min_{\beta \in [0,1]^m : \sum_{i \in M} \beta_i = 1} \max_{(\hat{s}_i)_{i \in M} \in [0,1]^m} \overline{W}(\hat{s}; \beta : M)
\end{aligned}$$

The first equality holds because $\beta_{j(k)}^* > 0$ means that $j(k) \in \arg \min_{i \in N} v_i(\hat{s}_i)(1 - P(\hat{s}_i))$. The second inequality holds because $v_h(s) \geq 0$ for all $s \in (0, 1]$ and $\hat{s}_j(i) \geq \hat{s}_k$ for all $k \in \mathcal{P}(i)$, which weakly increases the principal's payoff as $w_k(s) < 0$ for all $s \in (0, 1]$.

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